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Seismoelectric Exploration

Theory, Experiments, and Applications

Editors

Niels Grobbe

André Revil

Zhenya Zhu

Evert Slob

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FOREWORD 1

The topic of seismoelectric and electroseismic exploration that this book describes so completely is near and dear to my heart even if I have not been working in the area for some time now. Let me recall here some fondly-remembered anecdotes from my early connection to this field of research.

In 1991, when I was a young post-doc working for Professor Nafi Toksöz at MIT's Earth Resources Laboratory, Nafi let me go to the annual Society of Exploration Geophysics meeting in Houston and I attended a talk given by Dr. Art Thompson of Exxon Mobil Research entitled "Electroseismic Prospecting". Art described his beautifully-conceived field experiments where his team had sent seismic waves into the Earth from a source on the surface and recorded electric fields as the voltage difference between metal rods inserted into the ground. The electric fields seemed to be contained within the seismic waves themselves but also created from dipoles that were generated whenever the spherical seismic wave passed through an interface at depth. I was fascinated. When I was in graduate school at Texas A&M, in a class taught by Professor Dale Morgan called "Rock Physics", Dale (who is now at MIT) had described electrokinetic phenomena in porous media. It was intriguing to learn that fluid flow through the pores of a rock could drag along the extra cations of the nanoscale charge separation at the fluid-solid interface to produce a macroscale and easily measurable electric field. As a term paper for that class, I worked out how the electric field generated by the pressure gradients in a seismic wave influences the relative fluid-solid movement and the associated seismic attenuation. I later chose to make the theoretical understanding of seismic waves in porous media my PhD thesis topic. One of the chapters in the dissertation, and my first publication in 1991, focused on the electric fields generated within a seismic wave by electrokinetic effects.

So after attending Art's pioneering talk at the 1991 SEG meeting, I reported to Nafi that I was in a position to quantitatively model the seismically-generated electric fields that Art had recorded. Nafi told me to invite Art to come give his talk at MIT, which Art agreed to do. After Art's presentation, Nafi introduced himself and, in classic Nafi style, immediately invited Art to come do a sabbatical at MIT, which led to Art spending a semester with us at the Earth Resources Laboratory in 1992. Nafi also said that I should write a seismoelectric research proposal to Basic Energy Sciences of the U.S. Department of Energy. So I typed up a proposal covering

both theory and experiments and sent it into the program manager Bill Luth and to my surprise it was accepted for full funding a month or two later. Nafi let me work with one of his brilliant doctoral students Matthijs Haartsen and before long Matthijs had a numerical code simulating the electric fields generated by seismic waves propagating through layered porous media. Although my research focus across my career, both then and now, has been on more purely mechanical topics of porous-media response, the year or so I spent at MIT developing the equations, theory and simulations of seismoelectric response will always be a highlight.

Let me share a final particularly fond memory from 1992 at the Earth Resources Laboratory. Art and I had been discussing various types of laboratory experiments to quantify seismic-to-electrical coupling and, as is Art's style, some of the experiments he was conceiving of were quite sophisticated and all were quite expensive. One morning after getting funding, I was explaining to Zhenya Zhu, who is one of the editors of the present book, the basic idea of an electric field moving along with a compressional wave as part of the material response. Zhenya has been the Earth Resources Laboratory lead experimentalist over many decades. At that time, although he had created many types of ingenious lab-scale seismic models, he had not done much in the way of electrical measurements. An hour or so after our discussion, I was walking by Zhenya's lab and he had a huge slab of rock sitting on his bench top that he was hitting with a hammer. Two metal wires had been inserted into the rock and Zhenya was recording their voltage difference on an oscilloscope. Each time he hit the rock, the seismic pulse would appear on the screen of the oscilloscope; he was measuring the co-seismic electric field we had just discussed! I went and got Art Thompson and we all had a good laugh with how quickly and simply (and at zero cost) Zhenya had begun to observe seismic-to-electric coupling in the lab.

Although our various publications surrounding the theory and simulation of seismoelectric and electroseismic response did not come out until after I left MIT for the University of Paris (Institut de Physique du Globe), it was the nexus created by Nafi's management style at the Earth Resources Laboratory, where Art Thompson, Matthijs Haartsen, Zhenya Zhu and I were given the freedom to overlap and interact, that allowed this field to develop rapidly and in different directions in the early 1990s. I also recall Andre Revil, another editor of the present book, regularly dropping by my office at the Institut

de Physique du Globe de Paris in the mid 1990s. He was a graduate student at the University of Strasbourg at the time and very interested in seismoelectric and electrokinetic topics and we had many detailed discussions surrounding the various technical issues.

What makes the seismoelectric topic so fascinating is that the macroscopic response that is being recorded is generated from transport processes occurring within the electric double layer, which is typically a nanometer or less in width depending on the electrolyte salinity. Having response play out over so many orders of length scale and having the theoretical predictions be able to match the macroscale recordings (at least in some cases) represents everything that I find enjoyable about the study of multi-scale Earth materials. The present volume is a needed opportunity to shine light on this interesting and still developing field that ultimately should find applications even beyond the Earth sciences. What makes this book special is its coverage of all aspects of seismic-and-electrical coupling. From its nanometer-scale and

fluid-chemistry-dependent origins associated with the electric double layer, to a derivation of the pertinent macroscale governing equations and the associated modeling of the seismic and electrical sources, to laboratory studies, numerical simulation procedures and field studies, this book covers the current state of the art of seismoelectric and electroseismic exploration and is the essential reference for anybody interested in the topic.



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30 July, 2019

FOREWORD 2

Seismoelectricity is a ubiquitous natural phenomenon in the Earth with great potential to be utilized for scientific studies and for geophysical exploration. Put in the simplest terms, seismoelectricity occurs because seismic waves propagating through a porous, permeable medium induce fluid motion in the pores. The fluid contains ions; the ions in the fluid separate. One group of charged ions adheres to mineral surfaces around the pore wall leaving an access of free ions in the fluid. Seismic waves that induce the motion of the fluid also move the free ions and produce electrical current. The seismically induced “streaming current” is the source of seismoelectricity.

The seismoelectric phenomenon is fascinating both for its scientific significance but also for its potential applications to the Earth. Fluids play an important role in Earth’s crust, especially for determining its electrical properties. Seismic waves are most sensitive to elastic properties but not sensitive to fluid properties in the rocks. Seismic and electrical measurements, made separately, as is done in general practice, provide good information but cannot be correlated with the desired spatial resolution. Seismoelectrical measurements that combine both of these together focus directly to the source of conversion.

Seismoelectricity is a very broad field of study and it extends over physical, mathematical and earth sciences. Seminal papers in this field have been published in a wide spectrum of journals. This book, “Seismoelectric exploration: Theory, experiments and applications,” is a major contribution to the field. It provides a comprehensive overview of the seismoelectric phenomena. It contains 28 carefully selected articles representing a tremendous amount of work done by more than 60 authors. The papers are presented in four sections that cover the physical and theoretical aspects of seismoelectricity, laboratory measurements, numerical models and field data examples from earth observations.

The first section is on the theory. It includes the micro-scale origin of seismoelectric effects, electrokinetics, and streaming potential and the governing set of coupled seismo-electromagnetic equations. This section also includes the Green’s functions for earthquake sources.

The twelve papers in Section 2 represent an impressive number of laboratory studies on various aspects of seismoelectricity. They include measurement of streaming potentials, seismoelectric coupling coefficients, effects of

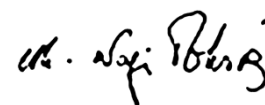
porosity, permeability anisotropy on seismoelectric conversion. Laboratory measurements in scaled-down bore-hole models are also included.

Section 3 is devoted to the modeling of seismoelectric signals due to various seismic sources including surface sources and earthquakes. Both point sources and finite rupture models are used for modeling. Seismoelectric conversion at interfaces and scattering at heterogeneities are also included.

The Section 4 “Field experiments and applications” is an important part of the book. It extends seismoelectricity from purely theoretical, numerical and laboratory domains into the real earth field applications. The papers in this section cover topics such as design of field instrumentation, noise removal and data processing. Field examples include measurements in unconsolidated sediments for shallow exploration, studies of ice sheets and glaciers, and seismoelectric signals related to earthquakes.

The editors have done an excellent job in their selection of the 28 papers in this book from a field of thousands of articles. Many of these papers are cited in the references. I would like to take the liberty of listing a few references that deserve special attention at the end of this Foreword.

The editors of “Seismoelectric Exploration: Theory, Experiments and Applications,” Niels Grobde, André Revil, Zhenya Zhu and Evert Slob must be congratulated for producing such a comprehensive volume on this important subject. The book will be an excellent resource to faculty and graduate students and researchers in academia, industry and government agencies. The scope of electroseismicity is far broader than geophysics and earth sciences. It could be applicable in any field dealing with fluids in porous media that can be imaged with acoustic and electromagnetic waves. Such imaging is widely used in many fields including medicine.



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12 August 2019

PREFACE

The aim of this book is to provide a wide overview of recent research activities on the coupling between poroelastic and electromagnetic disturbances, caused by charged particles in pore fluids. With this overview we hope to make the potential benefits of the phenomenon for several possible subsurface applications better known and more appreciated by geoscientists and engineers. We do this by adopting an approach that has a sound theoretical basis from which we explain which effects the coupling has and how it can provide more information on subsurface hydraulic, mechanical and electric properties compared with other more established geophysical methods.

The seismo-electromagnetic method is not well developed and is not widely applied in the field at the moment. In theory it can provide information on important porous medium properties such as water saturation, total dissolved solids, fractures, fluid properties, porosity, and permeability, that are of interest to environmental and engineering applications as well as for solid and fluid resource exploration and production. Two problems need to be overcome to make the method more widely accepted and used. The first is to understand how to separate the co-seismic, or co-electromagnetic, signal from the seismic to electromagnetic, and vice versa, conversions at boundaries where the medium properties have sudden changes. The second is to be able to measure quite weak signals with sufficient precision that relevant confidence bounds can be given to make the measured signals useful for analysis.

This book gives a thorough background of the underlying physics of the phenomenon, describes the different

models available, discusses various laboratory experiments, numerical modeling, field equipment and measurements, as well as provides an overview of applications, from exploration to cryosphere, well logging, and earthquake-related studies. This book is written for students, practitioners, and researchers in academia, industry, and agencies. Much of the material gathered in this book has been either presented at conferences, workshops and meetings, or published in journal papers, and has been further expanded and adjusted (including symbolic notations) to allow for optimal and coherent integration in the format of this book. It is an outgrowth of a dedicated technical session at the 2014 AGU Fall Meeting. Over the last years, more than 60 authors from three different continents have contributed to making this book a coherent overview of all aspects of coupling between elastic waves and electromagnetic fields in fluid-filled porous media. Almost all material can be used in courses on geophysical methods in undergraduate and graduate to postgraduate courses.

We hope you will enjoy studying this book as much as we enjoyed developing it, and that you will get inspired by the fascinating coupled seismo-electromagnetic phenomenon and its world of applications.

We thank AGU and Wiley for making this project possible and for having the patience to seeing it come to light.

**Niels Grobbe,
André Revil,
Zhenya Zhu,
and Evert Slob**

Part I: Theory

The Microscale Origin of the Seismoelectric Effect: Electrokinetics, Streaming Potential

Niels Grobbe^{1,2} and André Revil³

ABSTRACT

In this chapter, we discuss the microscale origin of coupled seismo-electromagnetic fields. We introduce the electrical double layer (EDL) and important electrokinetic processes that take place in porous materials that have a certain degree of pore-fluid saturation. The electrical double layer at the microscale plays a crucial role in the seismoelectric and electroseismic coupling processes. As a consequence of the presence of an EDL, several electrokinetic phenomena can occur, including the electro-osmotic phenomenon, where an electric field drives fluid flow, and the streaming potential phenomenon, where a hydraulic gradient creates an electric field. The streaming potential mechanism lies at the heart of the seismoelectric effect (mechanical to electromagnetic coupling). Electro-osmosis results in the so-called electroseismic effect (electromagnetic to mechanical coupling). For seismoelectric (and electroseismic) phenomena, we can distinguish three types of coupling: coseismic fields, interface response fields (or seismoelectric conversion), and source-converted fields. We introduce the so-called seismoelectric coupling coefficient, which describes the amount of coupling that naturally occurs between mechanical and electromagnetic fields in porous media.

1.1. INTRODUCTION

The use of seismo-electromagnetic and electromagnetic-seismic coupling effects are quite novel in geophysics. These types of coupling are one class of so-called electrokinetic effects in porous media in a broad sense (including colloidal suspensions). These coupling effects in the constitutive equations at the macroscale (i.e., at a representative elementary volume) are intrinsically

associated with the existence of an electrical double layer at the surface-interface between solid particles and pore water. These phenomena have been extensively studied throughout the years (e.g., Lyklema, 2002, 2003). They have a long history outside the realm of geophysics. For instance, Reuss (1809) was the first to prove the existence of such an electrokinetic effect. He performed a U-tube experiment demonstrating that the application of an electric current to a clay-sand-water mixture could cause the water to rise (the so-called electro-osmosis effect).

To understand electrokinetic phenomena in the context of geophysical applications, we consider a porous material such as a rock or soil. For the sake of simplicity, we first consider this material to be fully water saturated by an electrolyte. The surface of the grains (assumed to be insulating) is covered by reactive sites, such as silanol and aluminol sites for aluminosilicates. For some clays such as smectite, the mineral can also bear a charge through isomorphic substitutions in the crystalline

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framework (Davis et al., 1978; Avena and De Pauli, 1998; Revil et al., 2012).

The chemical reactions between the mineral surface and the pore water are collectively assembled in a so-called speciation model for a given mineral in contact with a specific pore water solution. These chemical reactions generate a net charge on the mineral surface depending on the presence and concentration of so-called potential-determining ions (e.g., Avena & De Pauli (1998) for clays, Heberling et al. (2014) for calcite). For alumino-silicates, these potential-determining ions are essentially H^+ and OH^- , because the surface sites are basically weak acids. Some ions from the pore water adsorb as inner- and outer-sphere complexes on the mineral surface, forming a so-called Stern layer (Stern, 1924; Davis et al., 1978). In this way, the Stern layer partially compensates the surface charge.

The surface charge that is not compensated in the Stern layer is compensated in the so-called diffuse layer. In this diffuse layer, the counterions and co-ions obey Boltzmann statistics. In other words, at equilibrium, the flux of ions due to the Coulombic field created by the surface charge counterbalances the diffusive flux of

ions (Gouy, 1910; Chapman, 1913; Revil & Jardani, 2013). As a result, the potential of the diffuse layer more or less exponentially decreases away from the grain surface toward the bulk of the pore space. As a side note, an exact exponential decay for the electrical potential in the diffuse layer (in absence of a macroscopic electrical field) is the result of linearizing the Poisson-Boltzmann equation, obtained by combining Gauss' law with the Boltzmann model of concentration in the Coulombic field created by the mineral surface charge (including the charge density of the Stern layer).

When the grains and the pore water are moving, the place of zero relative velocity is called the *shear plane*. This shear plane occurs somewhere in the diffuse layer. For the sake of simplicity, it is often considered to be at the interface between the Stern and diffuse layer. The electrostatic potential on this plane is referred to as the *zeta potential*. The Stern and diffuse layers together form the so-called electrical double layer (EDL). The characteristic length scale corresponding to half of the diffuse layer thickness is referred to as the *Debye length*, and is for typical reservoir rocks on the order of a few tens of nanometers

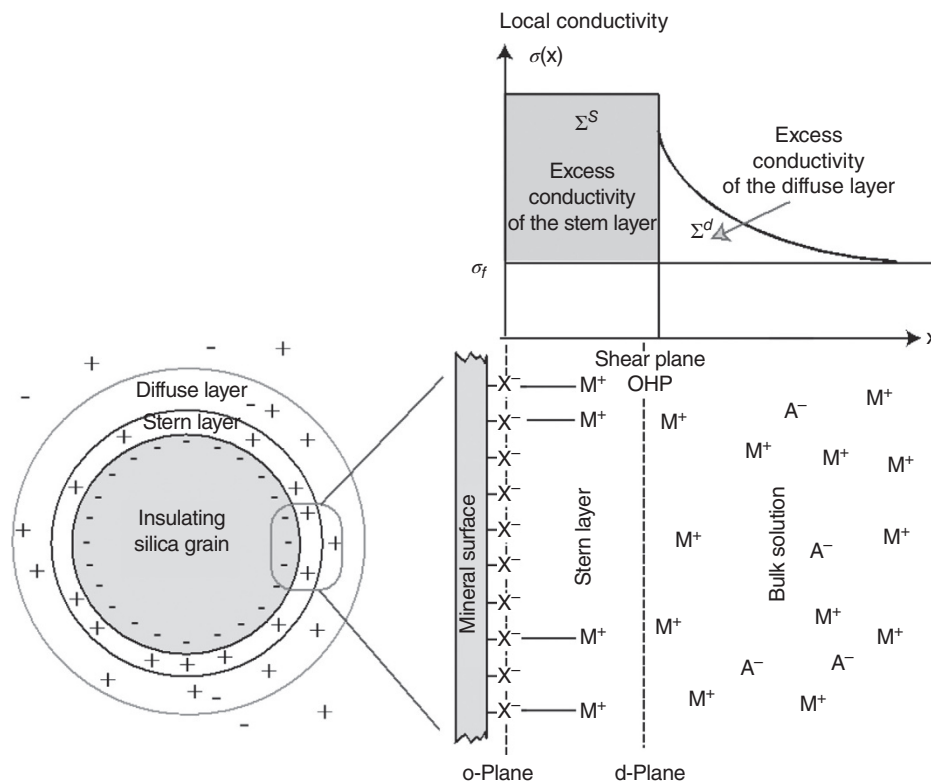


Figure 1.1 Sketch explaining the electrical double layer (EDL), which forms around the grain surfaces in a porous medium saturated with a fluid. The chemical speciation between the mineral and the pore water generates here a negative charge on the mineral surface. This negative charge is partially compensated by an excess of charge in the Stern layer (we do not distinguish here between inner and outer sphere complexes). Charge balance is obtained by the formation of a diffuse layer around the grain in which the charge concentration obeys Boltzmann statistics. The d-Plane is defined at the plane separating the diffuse layer from the Stern layer and is, therefore, the inner plane of the diffuse layer. The shear plane is assumed to be the same as this d-Plane. The o-Plane is the surface plane.

(Schoemaker et al., 2012). The above is schematically displayed in Figure 1.1.

Revil and Mahardika (2013) further distinguish two end-member scenarios, based on the size of the EDL with respect to the size of the pores:

1. A thick double layer, where the counterions of the diffuse layer are uniformly distributed in the pore space,
2. A thin double layer, where the thickness of the diffuse layer is much smaller than the size of the pores.

These end-member scenarios have an impact on the theoretical formulations and expressions for important petrophysical parameters such as the seismoelectric coupling coefficient, the permeability, and the electrical conductivity. These implications are further discussed in section 1.2. Note that throughout this book, we use the following convention

$$\hat{f}(x_i, \omega) = \int_{-\infty}^{\infty} f(x_i, t) e^{-j\omega t} dt, \quad (1.1)$$

for the definition for the forward temporal Fourier transform and

$$\tilde{f}(k_1, k_2, x_3, \omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \hat{f}(x_i, \omega) e^{jk_\alpha x_\alpha} dx_1 dx_2, \quad (1.2)$$

for the forward spatial Fourier transform. Here, ω denotes the angular frequency in $[\text{rad s}^{-1}]$, t is time, j indicates the imaginary unit, x_i indicates the three spatial directions of the right-handed Cartesian coordinate system (with the positive x_3 direction pointing downward [depth]), and k_α , with $\alpha = 1$ or 2 , denotes the horizontal wavenumber in those respective directions. The hat indicates a quantity in the space-frequency domain, the tilde sign denotes a quantity in the horizontal wavenumber-frequency domain. Throughout this book, Latin subscripts can take the values 1, 2, and 3, corresponding to the three spatial coordinate principal directions, whereas Greek subscripts can take the values 1 and 2. The Einstein summation convention holds for repeated indices (unless indicated otherwise).

As a consequence of the presence of an EDL, several electrokinetic phenomena can occur, including the electro-osmotic phenomenon, where an electric field drives fluid flow, and the streaming potential phenomenon, where a hydraulic gradient creates an electric field.

Helmholtz (1879) derived a mathematical description for electro-osmosis and electro-phoresis. The Helmholtz-Smoluchowski equation is one of the models used to describe such phenomena, and relates the streaming potential to controlling parameters such as the zeta-potential and fluid viscosity:

$$C_{s0} = \frac{\Delta\psi}{\Delta P} = \frac{\varepsilon_0 \varepsilon_r^f \zeta^p}{\eta \sigma^f}, \quad (1.3)$$

where C_{s0} denotes the steady-state streaming potential coefficient and $\Delta\psi$ is the induced electrical potential

difference (streaming potential) as a consequence of the pore-pressure difference ΔP .

Furthermore, ε_0 and ε_r^f are the electric permittivity of vacuum and the relative electric permittivity of the fluid, respectively, ζ^p represents the zeta-potential, η is the dynamic fluid viscosity, and σ^f denotes the fluid conductivity (Helmholtz, 1879; von Smoluchowski, 1903; Jouniaux & Bordes, 2012).

The streaming potential mechanism lies at the heart of the seismoelectric effect (mechanical to electromagnetic coupling). Electro-osmosis results in the so-called electro-seismic effect (electromagnetic to mechanical coupling). Onsager (1931a,b) (see also Lyklema, 2003), recognized reciprocity between streaming potential and electro-osmosis. This microscale reciprocity transfers to the macroscale, as long as the governing equations can be linearized, resulting in the seismoelectric and electroseismic effect to be reciprocal as well. This reciprocity in terms of macroscopic wave/field phenomena allows for geophysical imaging principles that exploit reciprocity principles, like interferometry, which can be applied to seismic, electromagnetic, and seismoelectric systems (Wapenaar & Fokkema, 2004, 2006; Bakulin & Calvert, 2006; Curtis et al., 2006; Schuster, 2009; De Ridder et al., 2009; Schoemaker et al., 2012). Seismoelectric signals versus electroseismic signals may sense the subsurface differently, offering different subsurface information (e.g., Thompson & Gist, 1993). Both coupling phenomena are attractive for exploration and monitoring purposes, and offer complementary information to classical seismic and electromagnetic techniques (e.g., Russell et al., 1997; Garambois & Dietrich, 2001; Revil et al., 2015; Zhu et al., 2016; Grobber & Slob, 2016).

To understand how the streaming potential mechanism results in the seismoelectric effect, we again revisit the EDL of Figure 1.1. In absence of flow, the porous system is electrically neutral. However, when a mechanical disturbance takes place, such as a seismic wave propagating through the porous formation, the streaming current phenomenon occurs. For example, a seismic pressure wave (P-wave), compresses and dilates the rock it passes. This compression (or dilation) creates a fluid pressure gradient within the seismic pulse, which induces pore fluid flow (e.g., Pride, 1994; Schoemaker et al., 2012; Revil et al., 2015). This fluid flow transports the excess electrical charge in the diffuse layer of the EDL, creating a streaming electric current (the streaming potential phenomenon).

We refer to this first type of coupling as the coseismic wave, since the coupling occurs at the location of the seismic wave at that instance in time, creating an electric field that is contained within the seismic wave.

When this wave encounters a heterogeneity in medium properties, for example an interface separating two different rock formation layers, or an arbitrarily shaped heterogeneity of sufficient size to be sensed by the wave,

asymmetry in the charge separation occurs, creating an independently electromagnetic field (with support outside the seismic wave) that theoretically resembles the description of an oscillating dipole (Garambois & Dietrich, 2002). This is an independently diffusing electromagnetic field, which, on a seismic timescale, arrives more or less instantaneously at the one-way seismic time (the time it takes for the seismic wave from the source to the heterogeneity generating the seismoelectric conversion). We refer to this response as the interface response, or seismo-electric conversion, and it is typically of greater interest due to the ability to investigate targets at depth.

A third type of seismoelectric coupling is sometimes distinguished: Seismo-electromagnetic conversion occurring directly upon source impact, generating localized charge asymmetries. Alternatively, this source-related coupling

can be considered a special case of the previously discussed interface response fields. Figure 1.2 schematically illustrates the three types of seismoelectric coupling that can occur, and their recordings by geophones and electrodes (seismogram and seismo-electrograms, respectively).

Intuitively, it may be more challenging to understand the reciprocal, electroseismic coupling. We assume a time-varying electric current injected into the ground, creating an electric field in the Earth. Through electro-osmosis, this electric field generates fluid flow.

Heterogeneities, such as an interface between two rock formations, will impact the electro-osmotic processes, creating a difference in how much fluid is driven into the interface from one side and the amount that is removed from the other side. This results in fluid

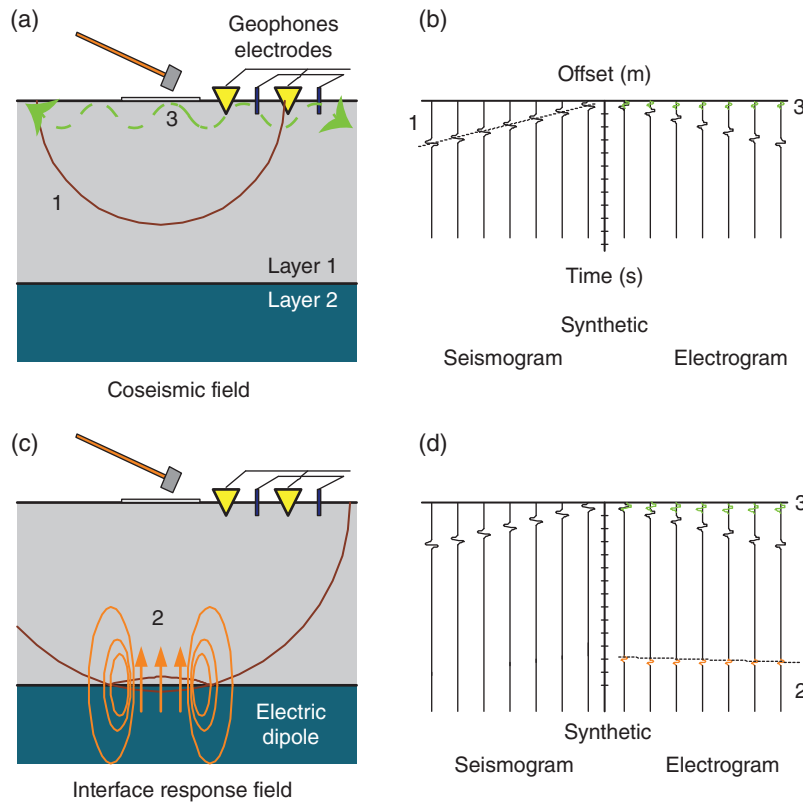


Figure 1.2 Schematic illustration of the three types of seismoelectric coupling: (a) coseismic fields, (c) interface response fields, and (b,d) source-converted diffusive EM fields. The left-hand side of the figure shows a cartoon of the effects, and the right-hand side shows how the different types of coupling would be measured by geophones (seismogram, left panel) and electrodes (electrogram, right panel). We see that the coseismic signals (event (1) in the electrogram) mimic the move-out behavior of the seismic wave in the seismogram (i.e., EM signals with a seismic velocity), whereas event 2 is present only in the electrogram and has more or less horizontal move-out characteristics, highlighting the much higher EM diffusive velocity per frequency for this interface response (or seismo-electric conversion) field. This event arrives at one-way seismic time, that is, the time it takes for the seismic wave to propagate from the location of the source to the interface/contrast generating the interface response field. From Grobbee (2016).

accumulation and dilation of the formation on one side, and fluid depletion and contraction on the other. The time-varying characteristics of such a mechanical dipole will generate an independent seismic wave (Pride, 2007; Thompson et al., 2007).

In both cases, we are dealing with frequency-dependent signals that are coupled.

The seismic wave creating the electrokinetic coupling contains a range of frequencies, causing frequency-dependent hydraulics and frequency-dependent streaming potential behavior (e.g., Reppert et al., 2001). Hence, instead of a steady-state streaming potential coefficient C_{s0} , one should consider frequency-dependent formulations for this coefficient (e.g., Packard, 1953; Pride, 1994; Reppert et al., 2001), since more charges can be dragged in an inertial laminar flow regime than in the viscous laminar flow regime (Pride, 1994; Revil & Mahardika, 2013). The transition between these different flow regimes is controlled by the so-called Biot critical frequency (this can also be described in terms of a characteristic relaxation time) or by the Reynolds number.

Pride (1994) derived a set of governing equations for the coupled seismo-electromagnetic phenomenon, which includes a frequency-dependent version of the streaming potential coefficient, formulated as a dynamic seismoelectric coupling coefficient $\hat{\mathcal{L}}$. A more extensive discussion of the governing equations is presented in section 1.2.

For seismic bandwidths, we can often make a low-frequency assumption, that is, considering frequencies smaller than the Biot critical frequency (e.g., Garambois & Dietrich, 2001). This has an impact on frequency-dependent medium parameters, such as $\hat{\mathcal{L}}$.

The low-frequency electrokinetic coupling $\mathcal{L}_0$, is related to the steady-state streaming potential coefficient via (e.g., Pride & Garambois, 2005):

$$\mathcal{L}_0 = -C_{s0}\sigma^e, \quad (1.4)$$

where $\sigma^e = \frac{\phi\sigma^f}{\alpha_\infty}$, with ϕ representing the porosity of the rock, and α_∞ , the tortuosity of the pore space. When we neglect effects from surface conductivity, combining equations 1.3 and 1.4 yields:

$$\mathcal{L}_0 = -\frac{\varepsilon_0 \varepsilon_r^f \zeta^p}{\eta F}, \quad (1.5)$$

where we have used the formation factor $F = \frac{\sigma^f}{\sigma^e}$ (this expression of the formation factor is only valid when surface-conductivity can be neglected and therefore be taken as a general definition of the formation factor).

Furthermore, the low-frequency assumption can be applied to the electromagnetic part of the governing equations, that is, we can use quasi-static formulations for the

Maxwell equations, making the seismoelectric description more similar to the theoretical descriptions of self-potential signals (e.g., Jardani et al., 2010). In these formulations, only the mechanical coupling with the electric field is considered, since at low frequencies, the electric and magnetic parts are not coupled (the electric field is rotation free). Consequently, we can write the electric field as:

$$\mathbf{E} = -\nabla\psi, \quad (1.6)$$

where $\mathbf{E}$ denotes the electric field vector, ∇ is the gradient operator, and ψ is the electrostatic potential. The rationale behind this approximation is that, besides the low-frequency bandwidth of the seismic signals, the targets of interest are sufficiently close to the receivers, such that we can neglect the time it takes for the EM converted fields to diffuse from the contrast generating the fields and the receivers recording them (e.g., Jardani et al., 2010). The quasi-static formulation of the seismoelectric phenomenon can help to reduce the degrees of freedom in the theoretical formulation of the problem. However, in the next chapter, we derive and describe the fully coupled seismo-electromagnetic system including induction effects.

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2

The Governing Set of Coupled Seismo-electromagnetic Equations

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ABSTRACT

We present the governing equations describing coupled seismo-electromagnetic wave propagation in porous media, based on Pride's assumptions. We start with the theory for fully saturated porous media, and reorganize the set of equations to allow for analytically based layered-Earth numerical modeling. We also present alternative seismo-electric equations based on Revil's assumptions. We finalize the first part of this chapter with a reciprocity numerical modeling example, using the layered-Earth modeling code ESSEMOD, for a model representing a configuration of a single layer over a halfspace, including a free surface. We display the results for the following source-receiver combinations: the electrical field component in the x_1 spatial direction due to a seismic dipole bulk force source in the same direction, and its reciprocal, the particle velocity field component in the x_1 spatial direction due to an electrical current source in the same direction. This example serves as a validation of the theoretical and numerical consistency. In the second part of this chapter, we describe the quasi-static approximation of the electromagnetic equations, and extend the seismoelectric theory toward partially saturated media, allowing us to study and describe the seismoelectric effect in media with different degrees of saturation. We also present the poroacoustic approximation of the seismoelectric equations, and subsequently expand and adopt this formulation to seismoelectric beamforming, or focusing. We illustrate the poroacoustic theoretical formulation and seismoelectric beamforming with a numerical example representing a water-flooding experiment containing varying degrees of saturation. This example shows the success of the seismoelectric beamforming approach in amplifying the seismo-electric conversion as well as the sensitivity of the seismoelectric converted signals to wetting/nonwetting contrasts.

2.1. INTRODUCTION

In Chapter 1 of this book, “The Microscale Origin of the Seismoelectric Effect: Electrokinetics, Streaming Potential,” we have discussed the microscale origin of the

coupled seismo-electromagnetic effect. We have seen that many microscale rock-fluid properties and interactions control the actual coupling that occurs between poroelastic-seismic wavefields and electromagnetic fields in porous media, having a certain degree of fluid saturation. We have also identified that these microscale rock-fluid interactions can give rise to three types of coupling.

1. The first type of coupling, occurs locally, inside the seismic waves or electromagnetic fields: the coupled fields travel along with the “mother” waves like a piggyback ride, inheriting their velocity. These coupled fields are referred to as coseismic or co-electromagnetic fields, respectively.

2. Second-order coupling, occurs when the waves encounter a porous medium heterogeneity / contrast in either mechanical, electromagnetic, and/or pore-fluid properties. Here, seismic energy is converted to

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electromagnetic energy, or vice versa, giving rise to converted fields. These fields are referred to as seismo-electromagnetic or electromagneto-seismic conversion, or also as interface response fields.

3. Type three, seismo-electromagnetic conversion, occurs directly at the source, due to localized charge asymmetries. This coupling can be considered a special case of the seismo-electromagnetic conversion.

In this chapter, we provide a set of macroscopic governing equations describing coupled seismo-electromagnetic wave propagation in porous media. We start with the theory for fully saturated porous media, and extend this theory toward partially saturated media, allowing us to study and describe the seismo-electromagnetic effect in media with different degrees of saturation. We discuss a variety of formulations of the seismo-electromagnetic phenomenon, starting with the formulation by Pride (1994). We then present alternative seismo-electromagnetic equations based on Revil's microscale electrical double layer models and discuss important underlying petrophysical principles and relationships, including the effects of the degree of saturation. We continue by discussing the Maxwell quasi-static approximation and its usability when studying the seismoelectric effect, and we then introduce a method to reduce the model complexity even further by considering an acoustic approximation.

After introducing the governing equations in their general form, we discuss how we can optimally reformulate and combine this set of equations for the specific scenario of a layered-Earth model. We discuss the possibility of separating the seismo-electromagnetic system in those scenarios into two independent propagation modes, the SH-TE propagation mode and the P-SV-TM propagation mode. In 2D models, this separation occurs naturally. In 3D scenarios, we have to define new mode quantities in order to carry out this separation. In the SH-TE mode, the horizontally polarized shear waves are coupled to the transverse electric mode, whereas in the P-SV-TM mode, the fast and slow Biot compressional waves are coupled to the transverse magnetic mode, via the vertically polarized shear waves. We express the vertical variations of the recorded field quantities for each propagation mode individually in terms of a system matrix containing the medium parameters and the horizontal partial differentiation operators acting on these field quantities. This reformulated set of equations can then form the basis for seismo-electromagnetic layered-Earth modeling (Woodhouse, 1974; Kennett, 1983; Ursin, 1983; Wapenaar & Grimbergen, 1996; Haartsen & Pride, 1997; Garambois & Dietrich, 2002; Ren et al., 2010; Hu & Gao, 2011; Grobbee & Slob, 2013; Grobbee et al., 2016 a,b).

The governing equations presented in this chapter form the foundation for all works presented in this book, each

of them using a certain degree of the theoretical formulations discussed here.

2.2. THEORY

2.2.1. Historical Overview

We start with a brief historical overview on the theoretical development of the seismo-electromagnetic phenomenon. We can distinguish two clear types of seismo-electromagnetic effects that have been studied and discussed in the past. Thompson (1936) postulated that an electric field change can occur associated to elastodynamic waves, since the mechanical disturbances change the local resistivity of the medium. He used a setup that actively generates a constant current between two electrodes and that measures the potential difference. While measuring, an elastodynamic wave propagates in the medium, passing by the electrodes and creating a change in potential difference. Thompson (1936) suggested that these measurements could be used to distinguish the propagation direction of different arrivals (for example horizontal versus vertical propagation). The second type of seismo-electromagnetic effect was first described by Ivanov (1940), who discusses the coupling between mechanical waves and fully coupled electromagnetic fields, where the electric and magnetic fields are influencing each other. This is the type of coupled seismo-electromagnetic phenomenon that we investigate in this book. A theory for wave propagation of electrokinetic phenomena in fluid-saturated porous media was developed by Frenkel (1944), in which he predicted the slow compressional wave and the seismo-electromagnetic effect. He made a marginal error in the development of the Biot-Gassmann constants and also only considered the electric effect and not the full set of Maxwell equations (Pride & Garambois, 2005). Martner and Sparks reported in 1959 that an electric potential difference generated by seismic wave propagation in the subsurface could be detected by electrode pairs (Martner & Sparks, 1959). A set of equations was then presented by Neev and Yeats (1989), in an attempt to extend Biot's theory to include the electrokinetic effects associated with the flow of fluid in poroelastic media. The theory was restricted to the resistive (low-frequency) domain in both fluid flow and electrical conduction. The Onsager reciprocity conditions enabled thermodynamic consistency. They did not use the Maxwell equations and also did not include the frequency-dependence of the transport equations. Using the principle of volume-averaging, Pride (1994) derived a set of governing equations describing the coupled seismo-electromagnetic system in fully saturated porous media, by coupling Biot's poroelasticity equations to the full Maxwell electromagnetic equations. This

theoretical model boosted the geophysical research on seismo-electromagnetic phenomena.

2.2.2. Governing Equations Based on Pride (1994)

We start by recapitulating the set of macroscopic governing equations for the coupled seismo-electromagnetic system in a fully fluid-saturated porous medium as derived by Pride (1994). We introduce some symbolic notational differences, we use an opposite definition of the Fourier transform, and we express the mechanical fields in terms of particle velocity fields instead of displacement fields (see also, Grobde, 2016).

In the coupled seismo-electromagnetic theory, we distinguish two equations of motion, two electromagnetic equations, and two stress-strain relationships. For notational convenience, we introduce in our notation the Laplace parameter $s = j\omega$.

2.2.2.1. Equations of Motion

The equations of motion derived by Pride (1994) are an extension of Biot's poroelastic equations of motion: they include a coupling term coupling the mechanical part to the electric field. This coupling physically represents an electro-osmotic feedback, describing the influence of the electromagnetic fields on the poroelastic wave propagation. Considering an arbitrary, inhomogeneous medium the two equations of motion including coupling are given by

$$s\rho^b \hat{v}_i^s + s\rho^f \hat{w}_i - \frac{1}{2} \frac{\partial (\hat{\tau}_{ij}^b + \hat{\tau}_{ji}^b)}{\partial x_j} = \hat{f}_i^b, \quad (2.1)$$

$$s\rho^f \hat{v}_i^s + \frac{\eta}{k} \hat{w}_i' + \frac{\partial \hat{p}^f}{\partial x_i} = \hat{f}_i^f, \quad (2.2)$$

where

$$\hat{w}_i' = \hat{w}_i - \hat{\mathcal{L}} \hat{E}_i, \quad (2.3)$$

$$\hat{w}_i = \phi (\hat{v}_i^f - \hat{v}_i^s), \quad (2.4)$$

$$\rho^b = \phi \rho^f + (1 - \phi) \rho^s. \quad (2.5)$$

In the above equations $\hat{w}_i$ is the Biot filtration velocity, $\hat{E}_i$ is the electric field strength, $\hat{\mathcal{L}}$ represents the dynamic seismo-electromagnetic coupling coefficient, which couples the elastodynamic wavefields to the electromagnetic wavefields, η represents the dynamic viscosity of the pore fluid, ϕ is the porosity of the medium, and $\hat{k}$ represents the frequency-dependent dynamic permeability. Combining these parameters yields the filtration velocity with coupling denoted by $\hat{w}_i'$. Furthermore, $\hat{v}_i^s$ and $\hat{v}_i^f$ are the solid and fluid particle velocities, respectively, $\hat{\tau}_{ij}^b$ represents the

bulk stress tensor, $\hat{p}^f$ is the fluid pressure, ρ^b , ρ^s , and ρ^f are the frequency-independent, isotropic bulk, solid, and fluid density, respectively, and $\hat{f}_i^b$ and $\hat{f}_i^f$ denote the volume densities of external force applied to the bulk and fluid phases, respectively.

The dynamic coupling coefficient $\hat{\mathcal{L}}$ can be written as follows:

$$\hat{\mathcal{L}} = \mathcal{L}_0 \left[1 + \frac{s}{\omega_c} \frac{m}{4} \left(1 - 2 \frac{d^l}{\Lambda} \right)^2 \left(1 - d^l \sqrt{\frac{s\rho^f}{\eta}} \right)^2 \right]^{-\frac{1}{2}}. \quad (2.6)$$

In equation (2.6), $\mathcal{L}_0$ represents the static coupling coefficient, which is defined as (Garambois & Dietrich, 2001; Pride & Garambois, 2005)

$$\mathcal{L}_0 = - \frac{\phi \varepsilon_0 \varepsilon_r^f \zeta^p}{\alpha_\infty \eta} \left(1 - 2 \frac{d^l}{\Lambda} \right). \quad (2.7)$$

Furthermore, ε_0 and ε_r^f are the electric permittivity of vacuum and the relative electric permittivity of the fluid, respectively. The symbol α_∞ represents the tortuosity of the porous medium. The tortuosity can be expressed in terms of the porosity and formation factor of the medium as $\alpha_\infty = \phi F$. The dynamic coupling coefficient has been experimentally investigated as a function of frequency (e.g., Schoemaker et al., 2012). In (2.6) and (2.7), Λ represents a pore throat parameter that can be seen as a weighted volume-to-surface ratio of the porous material and d^l the Debye screening length. The Debye screening length represents the characteristic thickness of the electrical double layer (EDL) and is typically on the order of a few to a hundred of nanometers (the latter under conditions of low ionic strength of the pore water). Physically, the Debye screening length can be considered as representing the distance over which mobile charge carriers, present in the electrolyte, screen out the surface charge present on the grains of the fluid-saturated porous medium (Debye & Hückel, 1923; Pride, 1994; Schoemaker et al., 2012), mathematically described as

$$d^l = \sum_{l=1}^L \sqrt{\frac{\varepsilon_0 \varepsilon_r^f k_B T}{e^2 z_l^2 N_l}}. \quad (2.8)$$

Here, k_B represents the Boltzmann constant, T the temperature (in [K]), N_l is the bulk ionic concentration of species l with valence z_l , for a total of L species present in the system, and e represents the elementary charge. We can calculate N_l using

$$N_l = 10^3 c N_A \|z_l'\|, \quad (2.9)$$

with N_A being Avogadro's constant, c the electrolyte concentration of the pore-fluid in [mol/L], and $\|z_l'\|$ being the

absolute value of the valency of the conjugate ion of zl. However, for realistic Earth scenarios, we typically consider the limit where $\frac{d^l}{\Lambda}$ is small, yielding

$$\mathcal{L}_0 = -\frac{\phi \epsilon_0 e_r^f \zeta^p}{\alpha_\infty \eta}. \quad (2.10)$$

Furthermore, ζ^p represents the zeta potential, an inner potential of the electrical diffuse layer coating the surface of the grains. Different expressions exist for the zeta potential. However, Pride and Morgan (1991) carried out a literature review on experimental studies of the zeta potential of quartz, showing that

$$\zeta^p = 8 + 26 \log_{10} c, \quad (2.11)$$

is a reasonable approximation of the zeta potential (in [mV]), where c denotes the electrolyte concentration of the pore fluid in [mol/L]. This equation was theoretically demonstrated and experimentally validated by Revil et al. (1999). It is important to realize that one may desire to use different zeta potential formulations, depending on the medium under consideration (Jackson et al., 2016). The zeta potential can be determined independently using an electrical double layer model combined with a speciation model for a given type of mineral (Leroy & Revil, 2004). Similar to the dynamic coupling coefficient, the dynamic permeability is defined as

$$\hat{k} = k_0 \left[\left(1 + \frac{s}{\omega_c} \frac{4}{m} \right)^{\frac{1}{2}} + \frac{s}{\omega_c} \right]^{-1}, \quad (2.12)$$

where, k_0 represents the static permeability. Under specific conditions, for example, constant grain size and tortuosity, the static permeability can be related to the porosity cubed using a Kozeny-Carman-type relationship, for example

$$k_0 = 2 \cdot 10^{-11} \phi^3, \quad (2.13)$$

for clean Fontainebleau sandstones (Bourbié et al., 1987). Furthermore, m is the similarity parameter, representing a combination of pore-geometry parameters

$$m = \frac{\phi \Lambda^2}{\alpha_\infty k_0}. \quad (2.14)$$

The similarity parameter is a dimensionless number, that will be fixed to a value of 8 (Johnson et al., 1987). Important to notice in both equations (2.6) and (2.12), ω_c represents the critical frequency separating the low-frequency viscous flow regime from the high-frequency inertial flow regime, defined as

$$\omega_c = \frac{\phi \eta}{\alpha_\infty k_0 \rho^f}. \quad (2.15)$$

For realistic rock and fluid parameters and general field scenarios, this transition frequency lies above the

maximum frequency of interest (this maximum being roughly between 10 and 1000 Hz). The frequency dependency of the dynamic coupling coefficient is related to relaxation mechanisms associated with the development of viscous boundary layers in the pores (Pride, 1994). So, when $\omega \ll \omega_c$, viscous boundary layers are not being developed yet, and hence we can neglect the second term of equation (2.6). This reduces the complex, frequency-dependent dynamic coupling coefficient to the static coupling coefficient

$$\hat{\mathcal{L}} = \mathcal{L}_0. \quad (2.16)$$

A similar reasoning holds for the dynamic permeability, which could be approximated by the static permeability for the frequency range under consideration. However, numerically speaking, coding the dynamic permeability is less tedious than coding the dynamic coupling coefficient, and therefore we use equation (2.12) in our numerical schemes.

Note that when the coupling coefficient is equal to zero, equations (2.1) and (2.2) reduce to Biot's poroelasticity equations and the elastodynamic waves decouple from the electromagnetic fields.

2.2.2.2. Rheology: Stress-Strain Relations

Similar to deriving the elastodynamic wave equation (Aki & Richards, 1980; Shearer, 2009), we need two stress-strain relations combined with the two equations of motion to be able to fully describe seismo-electromagnetic wave propagation in porous media. For an arbitrary, inhomogeneous, isotropic medium, the stress-strain relations are given by

$$-s \hat{\tau}_{ij}^b + d_{ijkl} \frac{\partial \hat{v}_k^s}{\partial x_l} + C \delta_{ij} \frac{\partial \hat{w}_k}{\partial x_k} = d_{ijkl} \hat{h}_{kl}^b + C \delta_{ij} \hat{q}^i, \quad (2.17)$$

$$s \hat{p}^f + C \frac{\partial \hat{v}_k^s}{\partial x_k} + M \frac{\partial \hat{w}_k}{\partial x_k} = C \hat{h}_{kk}^b + M \hat{q}^i, \quad (2.18)$$

with $\hat{h}_{ij}^b$ representing the density of external deformation rate acting on the bulk and $\hat{q}^i$ being the source volume density of injection rate in the fluid phase. The following symmetry properties hold: $\hat{\tau}_{ij}^b = \hat{\tau}_{ji}^b$, $\hat{h}_{ij}^b = \hat{h}_{ji}^b$ and $\hat{d}_{ijkl} = \hat{d}_{jikl} = \hat{d}_{iljk} = \hat{d}_{klji}$. Here, M and C are static scalar stiffness parameters and d_{ijkl} is a static stiffness tensor of rank four but, for isotropic media, a function of only two parameters given by

$$d_{ijkl} = \left[K_G - \frac{2}{3} G^f \right] \delta_{ij} \delta_{kl} + G^f (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (2.19)$$

where G^f denotes the shear modulus of the solid framework, K_G is Gassmann's bulk modulus and δ_{ij} denotes

the Kronecker delta. Isolating the divergence of the Biot filtration velocity $\frac{\partial \hat{w}_k}{\partial x_k}$ in (2.18), yields

$$\frac{\partial \hat{w}_k}{\partial x_k} = \frac{1}{M} \left[\hat{C} \delta_{kl} \hat{h}_{kl}^b + M \hat{q}^i - s \hat{p}^f - \hat{C} \delta_{kl} \frac{\partial \hat{v}_k^s}{\partial x_l} \right]. \quad (2.20)$$

We can define the stiffness parameters C and M as well as Gassmann's bulk modulus K_G in terms of the fluid, solid, and grain framework bulk moduli, yielding

$$K_G = \frac{K^f + \phi K^f + (1 + \phi) K^s \Delta}{1 + \Delta}, \quad (2.21)$$

$$C = \frac{K^f + K^s \Delta}{1 + \Delta}, \quad (2.22)$$

$$M = \frac{K^f}{\phi(1 + \Delta)}. \quad (2.23)$$

Here, K^s is the solid compression modulus, K^f the fluid compression modulus and K^{fr} the compression modulus of the solid framework. Δ is defined as

$$\Delta = \frac{K^f}{\phi(K^s)^2} [(1 - \phi) K^s - K^{fr}]. \quad (2.24)$$

Rewriting equations (2.17) and (2.20) for the isotropic, frequency-independent case results in

$$-s \hat{\tau}_{ij}^b + d_{ijkl} \frac{\partial \hat{v}_k^s}{\partial x_l} + C \delta_{ij} \frac{\partial \hat{w}_k}{\partial x_k} = d_{ijkl} \hat{h}_{kl}^b + C \delta_{ij} \hat{q}^i, \quad (2.25)$$

$$\frac{\partial \hat{w}_k}{\partial x_k} = \frac{1}{M} \left[\hat{C} \hat{h}_{kk}^b + M \hat{q}^i - s \hat{p}^f - C \delta_{kl} \frac{\partial \hat{v}_k^s}{\partial x_l} \right]. \quad (2.26)$$

Replacing $\frac{\partial \hat{w}_k}{\partial x_k}$ in (2.25) by the expression (2.26) eliminates this term, yielding

$$\begin{aligned} -s \hat{\tau}_{ij}^b + d_{ijkl} \frac{\partial \hat{v}_k}{\partial x_l} + \frac{C \delta_{ij}}{M} \left[C \delta_{kl} \hat{h}_{kl}^b + M \hat{q}^i - s \hat{p}^f - C \delta_{kl} \frac{\partial \hat{v}_k}{\partial x_l} \right] \\ = d_{ijkl} \hat{h}_{kl}^b + C \delta_{ij} \hat{q}^i. \end{aligned} \quad (2.27)$$

Bringing all source terms to the right-hand side and further simplifying yields

$$-s \hat{\tau}_{ij}^b + \left[d_{ijkl} - \frac{C^2}{M} \delta_{ij} \delta_{kl} \right] \frac{\partial \hat{v}_k}{\partial x_l} - \frac{sC}{M} \delta_{ij} \hat{p} = e_{ijkl} \hat{h}_{kl}^b, \quad (2.28)$$

with $e_{ijkl} = d_{ijkl} - \frac{C^2}{M} \delta_{ij} \delta_{kl}$. Alternatively, $e_{ijkl} = S \delta_{ij} \delta_{kl} + G^{fr} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$, where $S = K_G - \frac{2}{3} G^{fr} - \frac{C^2}{M}$, where we can recognize the elastodynamic Lamé constant $\lambda_c = K_G - \frac{2}{3} G^{fr}$. We can also work with the stiffness $H = K_G + \frac{4}{3} G^{fr}$ (e.g., Biot, 1962), which leads to the

alternative formulation $d_{ijkl} = (H - 2G^{fr}) \delta_{ij} \delta_{kl} + G^{fr} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$. Furthermore, we define $K_c = S + 2G^{fr}$ and we can relate $HM - C^2 = MK_c$.

2.2.2.3. Maxwell's Electromagnetic Field Equations With Coupling to Biot's Equations of Motion

Now that we have described the governing equations for the mechanical wave-part of the seismo-electromagnetic theory, we focus on the electromagnetic field-part. Below, Maxwell's equations for electromagnetic fields are presented, containing a coupling term to Biot's equations of motion in porous media (physically representing the streaming potential)

$$s \hat{D}_i + \hat{J}_i^{i,e} - \epsilon_{ijk} \frac{\partial \hat{H}_k}{\partial x_j} = -\hat{J}_i^e, \quad (2.29)$$

$$s \hat{B}_i + \hat{J}_i^{i,m} + \epsilon_{ijk} \frac{\partial \hat{E}_k}{\partial x_j} = -\hat{J}_i^m. \quad (2.30)$$

Here, $\hat{D}_i$ denotes the electric flux density and $\hat{B}_i$ the magnetic flux density. $\hat{E}_i$ and $\hat{H}_i$ represent the electric and magnetic field strength, respectively. The right-hand side of these two equations contains the external electric ($\hat{J}_i^e$) and magnetic ($\hat{J}_i^m$) current sources. Finally, $\hat{J}_i^{i,e}$ and $\hat{J}_i^{i,m}$ denote the averaged induced electric and magnetic current density, respectively. These induced current densities can be expressed as

$$\hat{J}_i^{i,e} = \hat{\sigma}^e \hat{E}_i - \hat{\mathcal{L}} \left(\frac{\partial \hat{p}^f}{\partial x_i} + s \rho^f \hat{v}_i^s - \hat{f}_i^f \right), \quad (2.31)$$

$$\hat{J}_i^{i,m} = \hat{\sigma}^m \hat{H}_i. \quad (2.32)$$

Here, $\hat{\sigma}^e$ and $\hat{\sigma}^m$ describe the frequency-dependent electric and magnetic conductivity, respectively. These two parameters describe electromagnetic relaxation. For Earth materials, we can ignore the magnetic relaxation losses, $\hat{\sigma}^m = 0$. Recognizing the expression for the equation of motion (2.2) in equation (2.31) and using it to express the induced electric current density in terms of the electric field and the filtration velocity yields

$$\hat{J}_i^{i,e} = \left(\hat{\sigma}^e - \frac{\eta}{\hat{k}} \hat{\mathcal{L}}^2 \right) \hat{E}_i + \frac{\eta}{\hat{k}} \hat{\mathcal{L}} \hat{w}_i. \quad (2.33)$$

In, equations (2.29) and (2.30), ϵ_{ijk} is the Levi-Civita symbol representing the alternating pseudo-tensor used to express (in combination with the spatial partial derivatives) the curls of the magnetic and electric fields, where

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } i, j, k \text{ is an even permutation of } 1, 2, 3, \\ -1 & \text{if } i, j, k \text{ is an odd permutation of } 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases} \quad (2.34)$$

The following constitutive relations hold:

$$\hat{D}_i = \varepsilon \hat{E}_i, \quad (2.35)$$

$$\hat{B}_i = \mu \hat{H}_i, \quad (2.36)$$

where $\varepsilon = \varepsilon_0 \varepsilon_r$ and $\mu = \mu_0 \mu_r$. Here, μ_0 representing the magnetic permeability of vacuum, μ_r represents the relative magnetic permeability, and ε_r the relative electric permittivity defined as

$$\varepsilon_r = \frac{\phi}{\alpha_\infty} (\varepsilon_r^f - \varepsilon_r^s) + \varepsilon_r^s, \quad (2.37)$$

where ε_r^s and ε_r^f are the relative electric permittivities of the solid and fluid, respectively. We choose $\mu_r = 1$, which is reasonable for Earth materials. In equation (2.33), one can clearly observe what happens with the coupling to Biot's poroelasticity equations when the coupling coefficient is chosen equal to zero.

When we substitute the constitutive relations (2.35) and (2.36) as well as equation (2.33) into Maxwell's electromagnetic field equations with coupling to Biot's equations of motion (2.29) and (2.30), the result reads

$$s\varepsilon \hat{E}_i + \left[\hat{\sigma}^e - \frac{\eta}{k} \hat{\mathcal{L}}^2 \right] \hat{E}_i + \frac{\eta}{k} \hat{\mathcal{L}} \hat{w}_i - \varepsilon_{ijk} \frac{\partial \hat{H}_k}{\partial x_j} = -\hat{J}_i^e, \quad (2.38)$$

$$s\mu_0 \hat{H}_i + \varepsilon_{ijk} \frac{\partial \hat{E}_k}{\partial x_j} = -\hat{J}_i^m. \quad (2.39)$$

Following Pride and Garambois (2005), neglecting surface conductivity, we can use the following reduced static expression (to leading order in $\frac{d^l}{\Lambda}$) for the frequency-independent electrical conductivity:

$$\hat{\sigma}^e = \frac{\phi \sigma^f}{\alpha_\infty}, \quad (2.40)$$

where we use the simplified version of the fluid conductivity for a binary symmetric electrolyte

$$\sigma^f = (ez)^2 N (b^+ + b^-). \quad (2.41)$$

This expression is based on the assumption that only one valency-type ion is present; therefore the subscript l is removed on z and N . Here, b^+ and b^- represent the ionic mobilities of the cations and anions, respectively. We use a typical value for the ionic mobility of $3.0 \cdot 10^{11} \text{ ms}^{-1} \text{ N}^{-1}$ for inorganic ions (Haartsen & Pride, 1997).

2.2.3. Governing Equations Based on Revil's Model

So far, we have discussed the seismo-electromagnetic theory following the assumptions of Pride (1994). Different underlying theoretical assumptions can lead to different expressions, especially for the dynamic coupling coefficient, the dynamic permeability, and the complex

conductivity. We now discuss the seismo-electromagnetic model according to Revil and Mahardika (2013). In this discussion, it is important to distinguish two end-member scenarios, based on the size of the electrical double layer with respect to the size of the pores (Revil & Mahardika, 2013):

1. A thick double layer, where the counterions of the diffuse layer are uniformly distributed in the pore space
2. A thin double layer, where the thickness of the diffuse layer is much smaller than the size of the pores

Pride (1994) makes use of a thin double layer theory based on the zeta potential. In contrast, Revil and Mahardika (2013) describe the (thick and thin) double layer theory based on the excess charge density. We here state the main alternative expressions for the dynamic coupling coefficient, permeability, and conductivity as presented by Revil and Mahardika (2013):

$$\underline{\hat{\mathcal{L}}}^*(\omega, s_w) = \frac{\underline{\hat{k}}^*(\omega, s_w) \hat{Q}_V^*(\omega, s_w)}{\eta_w}, \quad (2.42)$$

where the bar below the symbols indicates that it is a parameter according to Revil and Mahardika (2013), and the asterisk denotes that it is a complex parameter. For the low-frequency limit, we can replace these complex parameters by their low-frequency equivalents (removing the asterisk). Here, $\underline{\hat{\mathcal{L}}}^*(\omega, s_w)$ denotes the frequency- and saturation-dependent coupling coefficient, where s_w denotes the water saturation, and η_w the dynamic viscosity of the pore water. The complex frequency- and saturation-dependent permeability $\underline{\hat{k}}^*(\omega, s_w)$ is given (for our definition of the Fourier transform) by

$$\underline{\hat{k}}^*(\omega, s_w) = \frac{k_r k_0}{1 + j\omega\tau_k}, \quad (2.43)$$

and is used to describe the effect of the inertial force in Darcy's law. Here, τ_k is the relaxation time

$$\tau_k = \frac{k_r k_0 \rho_w F}{\eta_w} s_w^{1-n}, \quad (2.44)$$

characterizing the transition between the viscous laminar flow regime and the inertial laminar flow regime in the frequency-domain Navier-Stokes equation, where n is called the second Archie exponent. The critical angular frequency associated with this relaxation time is $\omega_c = \frac{1}{\tau_k}$.

Here, $F = \phi^{-m}$ is the formation factor that is related to the porosity by the first Archie law, where m is the cementation or porosity exponent (Archie, 1942). In addition, the relative permeability k_r can be used to express the relationship between the permeability and water saturation, using Brooks and Corey (1964)

$$k_r = s_w^{\frac{2+3n}{n}}, \quad (2.45)$$

where λ forms part of the exponent describing how permeability scales with water saturation, following a power-law behavior. It is shown in Revil (2013) that $\frac{2+3\lambda}{\lambda}$ can be written as $n+2$ to be the correct scaling factor when computing the permeability. One of the most important parameters in the model by Revil and Mahardika (2013) is $\hat{Q}(\omega, s_w)$, the frequency- and saturation-dependent excess of charge (or volumetric charge density) that can be dragged by the flow of the pore water through the pore space of the material. The frequency-dependency comes from the fact that more charges are dragged in the inertial flow regime than in the viscous laminar flow regime, which is in agreement with Pride (1994). We can further distinguish $\hat{Q}_V^0$ and $\hat{Q}_V^\infty$, representing the dynamic volumetric charge density dragged in the low-frequency ($\omega\tau_k \ll 1$; $\omega \ll \omega_c$) and high-frequency ($\omega\tau_k \gg 1$; $\omega \gg \omega_c$) regimes, respectively. The relaxation time (or similarly the critical frequency) distinguishes the different regimes. Therefore, we can use the following expression to compute the effective charge density as a function of frequency:

$$\frac{1}{\hat{Q}_V(\omega)} = \frac{1}{\hat{Q}_V^\infty} + \left(\frac{1}{\hat{Q}_V^0} - \frac{1}{\hat{Q}_V^\infty} \right) \frac{1}{\sqrt{1+j\omega\tau_k}}. \quad (2.46)$$

In the thick double layer approximation, $\hat{Q}_V^\infty \approx \hat{Q}_V^0$ (i.e., regardless of the frequency, all the counterions of the diffuse layer are dragged by the fluid flow), leading to the following saturation-dependent excess charge density:

$$\hat{Q}_V^*(\omega, s_w) \approx \frac{\hat{Q}_V^0}{s_w}. \quad (2.47)$$

For the thin double layer approximation it is expected that $\hat{Q}_V^\infty \gg \hat{Q}_V^0$ leading to

$$\hat{Q}_V^*(\omega, s_w) \approx \hat{Q}_V^0 s_w^{-1} \sqrt{1+j\omega\tau_k(s_w)}. \quad (2.48)$$

Another crucial difference between Pride (1994) and Revil and Mahardika (2013) lies in the formulation of the complex conductivity. The frequency-dependency of the complex conductivity can be directly attributed to the fact that the force applied to the charge carriers is controlled not only by the electric field, but also by an electrochemical potential. As a result, diffusion and electromigration are always coupled phenomena in porous media. As a second contribution to the total current density, we can distinguish the advective drag of excess charge of the pore space by the flow of the pore water. Following Revil and Mahardika (2013), we can express their complex conductivity $\hat{\sigma}^*(\omega)$ as

$$\hat{\sigma}^* = \sigma' - j\sigma'' = |\hat{\sigma}^*| \exp(-j\psi), \quad (2.49)$$

with the magnitude of the conductivity described by $|\hat{\sigma}^*| = \sqrt{(\sigma')^2 + (\sigma'')^2}$, and the phase lag between the electrical current and the resulting electrical field given by $\psi = \arctan \frac{\sigma''}{\sigma'}$. Here, σ' denotes the in-phase electrical conductivity, which can be expressed as a function of the pore water conductivity σ_w and the surface conductivity σ_s , as

$$\sigma' = \frac{1}{F} s_w^n \sigma_w + \sigma_s. \quad (2.50)$$

Expressions for the surface conductivity σ_s are discussed in Revil (2012, 2013). Furthermore, σ'' represents the quadrature conductivity, which can be expressed as

$$\sigma'' = -\frac{A(\phi, m)}{F} s_w^{n-1} b_s^+ \left(\frac{f_r}{1-f_r} \right), \quad (2.51)$$

with f_r representing the fraction of counterions in the diffuse layer, and b_s^+ denoting the mobility of the counterions in the Stern layer (Revil, 2012, 2013). The frequency-dependence of σ'' is not explicit in equation (2.51) (Revil, 2012). For the quasi-static scenario, this quadrature conductivity should go to zero under DC conditions. The quadrature conductivity typically becomes frequency-dependent below 0.1 Hz (Revil, 2012), a frequency that is below our seismo-electromagnetic range of interest. The above discussed complex conductivity agrees with experimental data. The complex conductivity as used in the model of Pride (1994), also contains frequency-dependency. However, this frequency-dependency is the result of an electro-osmotic contribution, which is expected to be negligible (Marshall & Madden, 1959). Pride (1994) ignores induced polarization effects. As a result of all this, the conductivity description based on the assumptions of Pride (1994) will not describe the observed quadrature conductivity of porous clayey materials correctly.

In the above, we have highlighted some important differences between the models of Pride (1994) and Revil and Mahardika (2013). Especially the formulation of Revil and Mahardika (2013) in terms of volumetric charge density and their definitions of the complex conductivity can be considered useful improvements. In addition, the theory as formulated by Revil and Mahardika (2013) allows for investigating seismo-electromagnetic effects in unsaturated porous media. Recently, Jardani and Revil (2015) extended the theory for full coupling in two-phase flow media. The theoretical description of partially saturated scenarios following the model of Pride (1994) has been presented in Warden et al. (2013). Despite obvious differences in the assumptions between the two models, in the low-frequency limit, which is the limit that holds

for most seismic scenarios (where the critical frequency lies above the bandwidth of interest), the model based on volumetric charge density as described by Revil and Mahardika (2013) is compatible with the model of Pride (1994), as shown in Appendix A of Revil and Mahardika (2013). The works presented in this book will use either the model of Pride (1994), or the models of Revil et al. as their foundations.

2.2.4. Reformulation of the Theory for Layered-Earth Models

Let us now consider the specific scenario of a layered-Earth geometry. In this scenario, the Earth is described as a layer-cake model (a package of different homogeneous layers), where the medium only varies in the vertical spatial direction (depth) and is considered to be constant in the horizontal spatial directions. For such a model, it makes sense to try and reformulate the theory, taking the vertical axis as the direction of preference and separating the vertical variations of the seismo-electromagnetic field quantities from the horizontal variations, allowing us to capture the seismo-electromagnetic system in the format of a general differential field equation (Wapenaar et al., 2008).

2.2.4.1. Separating Horizontal from Vertical Derivatives

We start with the following mechanical equations of motion (including electromagnetic coupling), stress-strain relations, and electromagnetic field equations in space-frequency:

$$s\rho^b \hat{v}_i^s + s\rho^f \hat{w}_i - \partial_j \hat{\tau}_{ij}^b = \hat{f}_i^b, \quad (2.52)$$

$$s\rho^f \hat{v}_i^s + \frac{\eta}{\bar{k}} \left(\hat{w}_i - \hat{\mathcal{L}} \hat{E}_i \right) + \partial_i \hat{p}^f = \hat{f}_i^f, \quad (2.53)$$

$$\partial_k \hat{w}_k = \frac{1}{M} \left(C \hat{h}_{kk}^b + M \hat{q}^i - s \hat{p}^f - C \partial_k \hat{v}_k^s \right), \quad (2.54)$$

$$-s \hat{\tau}_{ij}^b + e_{ijkl} \partial_l \hat{v}_k^s - \frac{sC}{M} \delta_{ij} \hat{p}^f = e_{ijkl} \hat{h}_{kl}^b, \quad (2.55)$$

$$s \epsilon \hat{E}_i + \left(\sigma^e - \frac{\eta}{\bar{k}} \hat{\mathcal{L}}^2 \right) \hat{E}_i + \frac{\eta}{\bar{k}} \hat{\mathcal{L}} \hat{w}_i - e_{ijk} \partial_j \hat{H}_k = -\hat{J}_i^e, \quad (2.56)$$

$$s\mu_0 \hat{H}_i + e_{ijk} \partial_j \hat{E}_k = -\hat{J}_i^m. \quad (2.57)$$

We now desire to separate the horizontal components from the vertical components. From equation (2.52) we obtain

$$\partial_3 \hat{\tau}_{i3}^b = s\rho^b \hat{v}_i^s + s\rho^f \hat{w}_i - \hat{f}_i^b - \partial_\alpha \hat{\tau}_{i\alpha}^b. \quad (2.58)$$

Using equation (2.54), we find

$$\begin{aligned} \partial_3 \hat{w}_3 &= \frac{1}{M} \left(C \hat{h}_{kk}^b + M \hat{q}^i - s \hat{p}^f - C \partial_k \hat{v}_k^s \right) - \partial_\alpha \hat{w}_\alpha \\ &= \frac{1}{M} \left(C \hat{h}_{kk}^b + M \hat{q}^i - s \hat{p}^f - C \partial_\alpha \hat{v}_\alpha^s \right) - \partial_\alpha \hat{w}_\alpha - \frac{C}{M} \partial_3 \hat{v}_3^s. \end{aligned} \quad (2.59)$$

Starting from equation (2.55), and separating the horizontal spatial derivatives acting on the particle velocity from the vertical derivatives, we find

$$e_{ijk3} \partial_3 \hat{v}_k^s = e_{ijkl} \hat{h}_{kl}^b + s \hat{\tau}_{ij}^b + \frac{sC}{M} \delta_{ij} \hat{p}^f - e_{ijka} \partial_\alpha \hat{v}_a^s. \quad (2.60)$$

We now select the x_3 -normal component $j = 3$ of the stress-tensor $\hat{\tau}_{ij}^b$, which is the only normal component that we desire in the final field quantity expressions. The other j -components of the stress tensor are eliminated, since they are not continuous over a horizontal interface, and since we are expressing the vertical variations of the field quantities in terms of their horizontal derivatives. We define the compliance tensor, being the inverse $\underline{e}_{p3i3}$ of the stiffness tensor e_{i3k3} , (remember that e_{ijkl} is an extension of the elastodynamic stiffness tensor d_{ijkl} , often also referred to as c_{ijkl}), as

$$\underline{e}_{p3i3} e_{i3k3} = \delta_{pk} \quad (2.61)$$

We apply this inverse to both the left-hand side and the right-hand sides of equation (2.60), leading to

$$\partial_3 \hat{v}_p^s = \underline{e}_{p3i3} \left(e_{i3kl} \hat{h}_{kl}^b + s \hat{\tau}_{i3}^b + \frac{sC}{M} \delta_{i3} \hat{p}^f - e_{i3ka} \partial_\alpha \hat{v}_a^s \right). \quad (2.62)$$

The compliance tensor is given explicitly by

$$\underline{e}_{p3i3} = \begin{pmatrix} \underline{e}_{1313} = \frac{1}{G^r} & \underline{e}_{1323} = 0 & \underline{e}_{1333} = 0 \\ \underline{e}_{2313} = 0 & \underline{e}_{2323} = \frac{1}{G^r} & \underline{e}_{2333} = 0 \\ \underline{e}_{3313} = 0 & \underline{e}_{3323} = 0 & \underline{e}_{3333} = \frac{1}{S + 2G^r} \end{pmatrix}. \quad (2.63)$$

Therefore, there are only nonzero solutions to equation (2.62) when $p = i$.

We focus on equation (2.58). We can recognize horizontal derivatives of the horizontal normal components of the stress tensor. As discussed, we only desire vertical normal components of the stress tensor in our final field equations. Therefore, using equation (2.55), we express these horizontal components of the stress tensor in terms of the other field components, as

$$\partial_\alpha \hat{\tau}_{i\alpha}^b = \partial_\alpha \frac{1}{S} \left(e_{iak\beta} \partial_\beta \hat{v}_k^s + e_{iak3} \partial_3 \hat{v}_k^s - \frac{sC}{M} \delta_{ia} \hat{p}^f - e_{iak\beta} \hat{h}_{k\beta}^b - e_{iak3} \hat{h}_{k3}^b \right). \quad (2.64)$$

We now rewrite equation (2.59), eliminating the vertical derivative of the vertical component particle velocity and the horizontal filtration velocity components using equations (2.62), with $k = 3$ and (2.66), respectively, yielding

$$\begin{aligned} \partial_3 \hat{w}_3 = & \frac{1}{M} \left(C \hat{h}_{kk}^b + M \hat{q}^i - s \hat{p}^f - C \partial_\alpha \hat{v}_\alpha^s \right) \\ & - \frac{C}{M} \left(\underline{e}_{3333} \left[e_{3311} \hat{h}_{11}^b + e_{3322} \hat{h}_{22}^b + e_{3333} \hat{h}_{33}^b + s \hat{\tau}_{33}^b \right] \right. \\ & \left. + \frac{sC}{M} \delta_{33} \hat{p}^f - e_{3311} \partial_1 \hat{v}_1^s - e_{3322} \partial_2 \hat{v}_2^s \right] \Bigg) \\ & - \left(-\frac{s \rho^f \hat{k}}{\eta} \partial_\alpha \hat{v}_\alpha^s + \hat{\mathcal{L}} \partial_\alpha \hat{E}_\alpha - \frac{\hat{k}}{\eta} \partial_\alpha \partial_\alpha \hat{p}^f + \frac{\hat{k}}{\eta} \partial_\alpha \hat{f}_\alpha^f \right). \end{aligned} \quad (2.65)$$

Similarly, from equation (2.53), we can find expressions for the horizontal components of the filtration velocity (since we only desire the vertical component in the final field quantity expressions), as

$$\hat{w}_\alpha = -\frac{s \rho^f \hat{k}}{\eta} \hat{v}_\alpha^s + \hat{\mathcal{L}} \hat{E}_\alpha - \frac{\hat{k}}{\eta} \partial_\alpha \hat{p}^f + \frac{\hat{k}}{\eta} \hat{f}_\alpha^f. \quad (2.66)$$

Combining equations (2.58) and (2.64), we substitute the expressions for the vertical derivative of the particle velocity (equation (2.62)) and the horizontal filtration velocity (equation (2.66)), to obtain for the x_1 -component of the x_3 -normal component of the bulk stress tensor:

$$\begin{aligned} \partial_3 \hat{\tau}_{13}^b = & s \rho^b \hat{v}_1^s + s \rho^f \left(-\frac{s \rho^f \hat{k}}{\eta} \hat{v}_1^s + \hat{\mathcal{L}} \hat{E}_1 - \frac{\hat{k}}{\eta} \partial_1 \hat{p}^f + \frac{\hat{k}}{\eta} \hat{f}_1^f \right) - \hat{f}_1^b \\ & - \frac{1}{s} \partial_\alpha \left\{ e_{1ak\beta} \partial_\beta \hat{v}_\alpha^s + e_{1ak3} \left(\underline{e}_{i3k3} \left(e_{i3mn} \hat{h}_{mn}^b + s \hat{\tau}_{i3}^b \right) \right. \right. \\ & \left. \left. + \frac{sC}{M} \delta_{i3} \hat{p}^f - e_{i3ma} \partial_\alpha \hat{v}_m^s \right) \right\} \\ & - \frac{sC}{M} \delta_{1a} \hat{p}^f - e_{1ak\beta} \hat{h}_{k\beta}^b - e_{1ak3} \hat{h}_{k3}^b \Bigg\}. \end{aligned} \quad (2.67)$$

We can observe that the term describing $e_{iak3} \partial - 0.0001 p t_3 \hat{v}_k^s$ only contributes if $k = 3$ (and hence $i = 3$), leading to

$$\begin{aligned} \partial_3 \hat{\tau}_{13}^b = & s \rho^b \hat{v}_1^s + s \rho^f \left(-\frac{s \rho^f \hat{k}}{\eta} \hat{v}_1^s + \hat{\mathcal{L}} \hat{E}_1 - \frac{\hat{k}}{\eta} \partial_1 \hat{p}^f + \frac{\hat{k}}{\eta} \hat{f}_1^f \right) - \hat{f}_1^b \\ & - \frac{1}{s} \partial_\alpha \left\{ e_{1ak\beta} \partial_\beta \hat{v}_\alpha^s + e_{1a33} \left(\underline{e}_{3333} \left(e_{33mn} \hat{h}_{mn}^b + s \hat{\tau}_{33}^b \right) \right. \right. \\ & \left. \left. + \frac{sC}{M} \delta_{33} \hat{p}^f - e_{33ma} \partial_\alpha \hat{v}_m^s \right) \right\} \\ & - \frac{sC}{M} \delta_{1a} \hat{p}^f - e_{1ak\beta} \hat{h}_{k\beta}^b - e_{1ak3} \hat{h}_{k3}^b \Bigg\}. \end{aligned} \quad (2.68)$$

In a similar way, choosing $i = 2$ in equations (2.58) and (2.64) and combining both equations, followed by

substitution of the expressions for the vertical derivative of the particle velocity (equation (2.62)) and the horizontal filtration velocity (equation (2.66)), we obtain the expression for $\partial - 0.0001 p t_3 \hat{\tau}_{23}^b$.

From equation (2.56) we obtain,

$$\partial_3 \hat{H}_1 = s \hat{E}_2 + \frac{\eta}{\hat{k}} \hat{\mathcal{L}} \hat{w}_2 + \partial_1 \hat{H}_3 + \hat{J}_2^e, \quad (2.69)$$

$$\partial_3 \hat{H}_2 = -s \hat{E}_1 - \frac{\eta}{\hat{k}} \hat{\mathcal{L}} \hat{w}_1 + \partial_2 \hat{H}_3 - \hat{J}_1^e, \quad (2.70)$$

where we made use of $\hat{e} = \varepsilon + \frac{\hat{\sigma}^e}{s} - \rho^E \hat{\mathcal{L}}^2$. Similarly, using equation (2.57), we derive

$$\partial_3 \hat{E}_1 = -s \mu_0 \hat{H}_2 + \partial_1 \hat{E}_3 - \hat{J}_2^m, \quad (2.71)$$

$$\partial_3 \hat{E}_2 = s \mu_0 \hat{H}_1 + \partial_2 \hat{E}_3 + \hat{J}_1^m. \quad (2.72)$$

Now explicitly evaluating $e_{ijkl} = S \delta_{ij} \delta_{kl} + G^{fr} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$, as well as using equation (2.63), yields

$$\partial_3 \hat{v}_1^s = \left(\hat{h}_{31}^b + \hat{h}_{13}^b \right) + \frac{s}{G^{fr}} \hat{\tau}_{13}^b - \partial_1 \hat{v}_3^s, \quad (2.73)$$

$$\partial_3 \hat{v}_2^s = \left(\hat{h}_{32}^b + \hat{h}_{23}^b \right) + \frac{s}{G^{fr}} \hat{\tau}_{23}^b - \partial_2 \hat{v}_3^s, \quad (2.74)$$

$$\begin{aligned} \partial_3 \hat{v}_3^s = & \frac{s}{K_c} \hat{\tau}_{33}^b + \frac{sC}{MK_c} \hat{p}^f - \frac{S}{K_c} (\partial_1 \hat{v}_1^s + \partial_2 \hat{v}_2^s) \\ & + \frac{S}{K_c} \left(\hat{h}_{11}^b + \hat{h}_{22}^b \right) + \hat{h}_{33}^b \end{aligned} \quad (2.75)$$

$$\begin{aligned} \partial_3 \hat{w}_3 = & -\frac{sC}{MK_c} \hat{\tau}_{33}^b - \frac{s}{M} \left(1 + \frac{C^2}{MK_c} \right) \hat{p}^f + \frac{\hat{k}}{\eta} \partial_\alpha \hat{p}^f \\ & + \frac{\rho^f}{\hat{\rho}^E} \partial_\alpha \hat{v}_\alpha^s - \hat{\mathcal{L}} \partial_\alpha \hat{E}_\alpha - \frac{\hat{k}}{\eta} \partial_\alpha \hat{f}_\alpha^f - \frac{C}{M} \left(1 - \frac{S}{K_c} \right) \partial_\alpha \hat{v}_\alpha^s \\ & - \frac{C}{M} \frac{S}{K_c} \hat{h}_{aa}^b + \frac{C}{M} \hat{h}_{aaa}^b + \hat{q}^i, \end{aligned} \quad (2.76)$$

$$\begin{aligned} \partial_3 \hat{\tau}_{13}^b = & -\partial_1 \frac{S}{K_c} \hat{\tau}_{33}^b - s \rho^f \frac{\hat{k}}{\eta} \partial_1 \hat{p}^f + \frac{1}{s} \partial_1 \left(\frac{sC}{M} \left[1 - \frac{S}{K_c} \right] \hat{p}^f - \right. \\ & \left. \left[\frac{4G^{fr}(S + G^{fr})}{K_c} \partial_1 \hat{v}_1^s + \frac{2SG^{fr}}{K_c} \partial_2 \hat{v}_2^s \right] \right) \\ & + s \rho^b \hat{v}_1^s - s^2 (\rho^f)^2 \frac{\hat{k}}{\eta} \hat{v}_1^s + s \rho^f \hat{\mathcal{L}} \hat{E}_1 - \hat{f}_1^b + s \rho^f \frac{\hat{k}}{\eta} \hat{f}_1^f \\ & - \frac{1}{s} \partial_1 \left(\frac{S^2}{K_c} \left[\hat{h}_{11}^b + \hat{h}_{22}^b \right] \right) + \frac{1}{s} \partial_1 \left(K_c \hat{h}_{11}^b + S \hat{h}_{22}^b \right) \\ & - \frac{1}{s} \partial_2 \left(G^{fr} [\partial_1 \hat{v}_2^s + \partial_2 \hat{v}_1^s] \right) + \frac{1}{s} \partial_2 \left(G^{fr} [\hat{h}_{21}^b + \hat{h}_{12}^b] \right), \end{aligned} \quad (2.77)$$

$$\begin{aligned}
\partial_3 \hat{\tau}_{23}^b = & -\partial_2 \frac{S}{K_c} \hat{\tau}_{33}^b - s\rho^f \frac{\hat{k}}{\eta} \partial_2 \hat{p}^f + \frac{1}{s} \partial_2 \left(\frac{sC}{M} \left[1 - \frac{S}{K_c} \right] \hat{p}^f - \right. \\
& \left. \left[\frac{2SG^{fr}}{K_c} \partial_1 \hat{v}_1^s + \frac{4G^{fr}(S + G^{fr})}{K_c} \partial_2 \hat{v}_2^s \right] \right) \\
& + s\rho^b \hat{v}_2^s - s^2 (\rho^f)^2 \frac{\hat{k}}{\eta} \hat{v}_2^s + s\rho^f \hat{\mathcal{L}} \hat{E}_2 - \hat{f}_2^b + s\rho^f \frac{\hat{k}}{\eta} \hat{f}_2^f \\
& - \frac{1}{s} \partial_2 \left(\frac{S^2}{K_c} [\hat{h}_{11}^b + \hat{h}_{22}^b] \right) + \frac{1}{s} \partial_2 (S \hat{h}_{11}^b + K_c \hat{h}_{22}^b) \\
& - \frac{1}{s} \partial_1 (G^{fr} [\partial_1 \hat{v}_2^s + \partial_2 \hat{v}_1^s]) + \frac{1}{s} \partial_1 (G^{fr} [\hat{h}_{21}^b + \hat{h}_{12}^b]).
\end{aligned} \quad (2.78)$$

From equation (2.58), we also directly obtain

$$\partial_3 \hat{\tau}_{33}^b = s\rho^b \hat{v}_3^s + s\rho^f \hat{w}_3 - \hat{f}_3^b - (\partial_1 \hat{\tau}_{31}^b + \partial_2 \hat{\tau}_{32}^b). \quad (2.79)$$

From equation (2.53), we derive

$$-\partial_3 \hat{p}^f = s\rho^f \hat{v}_3^s + \frac{\eta}{\hat{k}} (\hat{w}_3 - \hat{\mathcal{L}} \hat{E}_3) - \hat{f}_3^f. \quad (2.80)$$

Starting from equations (2.69) and (2.70), we eliminate $\hat{H}_3$ using

$$\hat{H}_3 = \frac{1}{s\mu_0} (\partial_2 \hat{E}_1 - \partial_1 \hat{E}_2 - \hat{J}_3^m), \quad (2.81)$$

and the horizontal filtration velocity components using equation (2.66), which leads to

$$\begin{aligned}
\partial_3 \hat{H}_1 = & s\hat{e} \hat{E}_2 + \frac{\eta}{\hat{k}} \hat{\mathcal{L}} \left(-\frac{s\rho^f \hat{k}}{\eta} \hat{v}_2^s + \hat{\mathcal{L}} \hat{E}_2 - \frac{\hat{k}}{\eta} \partial_2 \hat{p}^f + \frac{\hat{k}}{\eta} \hat{f}_2^f \right) \\
& + \frac{1}{s\mu_0} (\partial_1 \partial_2 \hat{E}_1 - \partial_1^2 \hat{E}_2 - \partial_1 \hat{J}_3^m) + \hat{J}_2^e,
\end{aligned} \quad (2.82)$$

$$\begin{aligned}
\partial_3 \hat{H}_2 = & -s\hat{e} \hat{E}_1 - \frac{\eta}{\hat{k}} \hat{\mathcal{L}} \left(-\frac{s\rho^f \hat{k}}{\eta} \hat{v}_1^s + \hat{\mathcal{L}} \hat{E}_1 - \frac{\hat{k}}{\eta} \partial_1 \hat{p}^f + \frac{\hat{k}}{\eta} \hat{f}_1^f \right) \\
& + \frac{1}{s\mu_0} (\partial_2^2 \hat{E}_1 - \partial_2 \partial_1 \hat{E}_2 - \hat{J}_3^m) - \hat{J}_1^e.
\end{aligned} \quad (2.83)$$

Finally, starting from equations (2.71) and (2.72), we eliminate $\hat{E}_3$ using

$$\hat{E}_3 = \frac{1}{s\hat{e}} \left(-\frac{\eta}{\hat{k}} \hat{\mathcal{L}} \hat{w}_3 + \partial_1 \hat{H}_2 - \partial_2 \hat{H}_1 - \hat{J}_3^e \right), \quad (2.84)$$

which yields

$$\partial_3 \hat{E}_1 = -s\mu_0 \hat{H}_2 + \frac{1}{s\hat{e}} \left(-\frac{\eta}{\hat{k}} \hat{\mathcal{L}} \partial_1 \hat{w}_3 + \partial_1^2 \hat{H}_2 - \partial_1 \partial_2 \hat{H}_1 - \partial_1 \hat{J}_3^e \right) - \hat{J}_2^m, \quad (2.85)$$

$$\partial_3 \hat{E}_2 = s\mu_0 \hat{H}_1 + \frac{1}{s\hat{e}} \left(-\frac{\eta}{\hat{k}} \hat{\mathcal{L}} \partial_2 \hat{w}_3 + \partial_2 \partial_1 \hat{H}_2 - \partial_2^2 \hat{H}_1 - \partial_2 \hat{J}_3^e \right) + \hat{J}_1^m. \quad (2.86)$$

We now have expressed the vertical derivatives of the desired field quantities in terms of other fields and the horizontal derivatives on other fields.

2.2.4.2. Combining Field Quantities into Mode Quantities

Since we are considering a medium that is invariant in two directions (layered-Earth), it can be useful to decouple the total 3D system of equations into two independent seismo-electromagnetic propagation modes: the SH-TE propagation mode, where the horizontally polarized shear-waves are coupled to the transverse electric mode, and the P-SV-TM propagation mode where the Biot fast and slow pressure waves are coupled to the transverse magnetic mode via the vertically polarized shear waves. To decouple the system in these two independent propagation modes, we first apply a 2D spatial Fourier transformation to equations (2.73)–(2.86) to obtain the horizontal wavenumber-frequency domain expressions (denoted by the tilde sign). This is possible, since we do not consider lateral medium variations. We can then express these equations in terms of their mode quantities, by taking the divergence or rotation of the expressions under consideration. To this end, let us first define the following mode quantities:

$$\tilde{v}^{s,H} = -jk_1 \tilde{v}_2^s + jk_2 \tilde{v}_1^s, \quad (2.87)$$

$$\tilde{\tau}^{b,H} = -jk_1 \tilde{\tau}_{23}^b + jk_2 \tilde{\tau}_{13}^b, \quad (2.88)$$

$$\tilde{E}^H = -jk_1 \tilde{E}_2 + jk_2 \tilde{E}_1, \quad (2.89)$$

$$\tilde{H}^H = -jk_1 \tilde{H}_1 - jk_2 \tilde{H}_2, \quad (2.90)$$

for the SH-TE mode, and

$$\tilde{v}^{s,V} = \frac{k_1}{\kappa} \tilde{v}_1^s + \frac{k_2}{\kappa} \tilde{v}_2^s, \quad (2.91)$$

$$\tilde{\tau}^{b,V} = \frac{k_1}{\kappa} \tilde{\tau}_{13}^b + \frac{k_2}{\kappa} \tilde{\tau}_{23}^b, \quad (2.92)$$

$$\tilde{E}^V = \frac{k_1}{\kappa} \tilde{E}_1 + \frac{k_2}{\kappa} \tilde{E}_2, \quad (2.93)$$

$$\tilde{H}^V = \frac{k_1}{\kappa} \tilde{H}_2 - \frac{k_2}{\kappa} \tilde{H}_1, \quad (2.94)$$

for the P-SV-TM mode. Here the radial wavenumber $\kappa = \sqrt{k_1^2 + k_2^2}$. For P-SV-TM, the subscript “norm” indicates that these specific mode quantities have been normalized with a factor $-j\kappa$ (this to obtain favorable symmetry properties in the system matrix later on). Consequently, the factors on the right-hand side, before the field quantities, have also been normalized by $-j\kappa$. The field quantities that are not present in the mode expressions above are purely P-SV-TM field quantities. We can observe for example in equation (2.87), that by taking jk_2 of $\tilde{v}_1^s$ (corresponding to $-\partial - 0.0001pt_2 \hat{v}_1^s$ and adding

this to $-jk_1$ of $\tilde{v}_3^s$) yields the SH-TE particle velocity mode quantity. Up to this point, we have eliminated $\hat{\tau}_{\alpha\beta}^b$, $\hat{w}_\alpha$, $\hat{H}_3$, and $\hat{E}_3$, using the horizontal wavenumber-frequency versions of equations (2.55), (2.66), (2.81), and (2.84), respectively, according to

$$\tilde{\tau}_{\alpha\beta}^b = -\frac{1}{s} \left(e_{\alpha\beta k l} j k_l \tilde{v}_k^s + \frac{sC}{M} \delta_{\alpha\beta} \tilde{p}^f + e_{\alpha\beta k l} \tilde{h}_{kl}^b \right), \quad (2.95)$$

$$\tilde{w}_\alpha = -\frac{s\rho^f \hat{k}}{\eta} \tilde{v}_\alpha^s + \hat{\mathcal{L}} \tilde{E}_\alpha + \frac{\hat{k}}{\eta} j k_\alpha \tilde{p}^f + \frac{\hat{k}}{\eta} \tilde{f}_\alpha^f, \quad (2.96)$$

$$\tilde{H}_3 = \frac{1}{s\mu_0} \left(-\tilde{E}^H - \tilde{J}_3^m \right), \quad (2.97)$$

$$\tilde{E}_3 = \frac{1}{s\hat{\epsilon}} \left(-\frac{\eta}{\hat{k}} \hat{\mathcal{L}} \tilde{w}_3 - j\kappa \tilde{H}_{norm}^V - \tilde{J}_3^e \right). \quad (2.98)$$

We now take the derivatives of equations (2.73)–(2.86) and combine according to equations (2.87)–(2.94), yielding two sets of governing equations, one for the SH-TE mode and one for the P-SV-TM mode, expressing the vertical variations of the mode and field quantities in terms of the horizontal variations. For the SH-TE mode, this results in

$$\partial_3 \tilde{v}^{s,H} = \frac{s}{G^{fr}} \tilde{\tau}^{b,H} - jk_1 \left(\tilde{h}_{23}^b + \tilde{h}_{32}^b \right) + jk_2 \left(\tilde{h}_{13}^b + \tilde{h}_{31}^b \right), \quad (2.99)$$

$$\begin{aligned} \partial_3 \tilde{\tau}^{b,H} &= \frac{G^{fr}}{s} \kappa^2 \tilde{v}^{s,H} + \left(s\rho^b - s^2 (\rho^f)^2 \frac{\hat{k}}{\eta} \right) \tilde{v}^{s,H} + s\rho^f \hat{\mathcal{L}} \tilde{E}^H + \\ &jk_1 \left(\tilde{f}_2^b - s\rho^f \frac{\hat{k}}{\eta} \tilde{f}_2^f \right) - jk_2 \left(\tilde{f}_1^b - s\rho^f \frac{\hat{k}}{\eta} \tilde{f}_1^f \right) \\ &+ \frac{G^{fr}}{s} \left(-k_1^2 + k_2^2 \right) \left(\tilde{h}_{21}^b + \tilde{h}_{12}^b \right) - k_1 k_2 \frac{2G^{fr}}{s} \left(-\tilde{h}_{11}^b + \tilde{h}_{22}^b \right), \end{aligned} \quad (2.100)$$

$$\partial_3 \tilde{E}^H = s\mu_0 \tilde{H}^H - jk_\alpha \tilde{J}_\alpha^m, \quad (2.101)$$

$$\begin{aligned} \partial_3 \tilde{H}^H &= s\hat{\epsilon} \tilde{E}^H - s\rho^f \hat{\mathcal{L}} \tilde{v}^{s,H} + \frac{\eta}{\hat{k}} \hat{\mathcal{L}}^2 \tilde{E}^H + \frac{1}{s\mu_0} \kappa^2 \tilde{E}^H \\ &+ \hat{\mathcal{L}} \left(-jk_1 \tilde{f}_2^f + jk_2 \tilde{f}_1^f \right) - jk_1 \tilde{J}_2^e + jk_2 \tilde{J}_1^e + \frac{1}{s\mu_0} \kappa^2 \tilde{J}_3^m. \end{aligned} \quad (2.102)$$

Similarly, for the normalized P-SV-TM mode, we obtain

$$\partial_3 \tilde{v}_{norm}^{s,V} = \frac{s}{G^{fr}} \tilde{\tau}_{norm}^{b,V} + j\kappa \tilde{v}_3^s + \frac{k_\alpha}{\kappa} \left(\tilde{h}_{\alpha 3}^b + \tilde{h}_{3\alpha}^b \right), \quad (2.103)$$

$$\partial_3 \tilde{v}_3^s = \frac{s}{K_c} \tilde{\tau}_{33}^b + \frac{sC}{MK_c} \tilde{p}^f + \frac{j\kappa S}{K_c} \tilde{v}_{norm}^{s,V} + \frac{S}{K_c} \left(\tilde{h}_{11}^b + \tilde{h}_{22}^b \right) + \tilde{h}_{33}^b, \quad (2.104)$$

$$\begin{aligned} \partial_3 \tilde{w}_3 &= -s \frac{C}{MK_c} \tilde{\tau}_{33}^b - \frac{s}{M} \left(1 + \frac{C^2}{MK_c} \right) \tilde{p}^f - \frac{\hat{k}}{\eta} \kappa^2 \tilde{p}^f \\ &- j\kappa \frac{\rho^f}{\hat{\rho}^E} \tilde{v}_{norm}^{s,V} + j\kappa \frac{C}{M} \left(1 - \frac{S}{K_c} \right) \tilde{v}_{norm}^{s,V} + j\kappa \hat{\mathcal{L}} \tilde{E}_{norm}^V \\ &+ \frac{\hat{k}}{\eta} jk_\alpha \tilde{f}_\alpha^f - \frac{CS}{MK_c} \left(\tilde{h}_{11}^b + \tilde{h}_{22}^b \right) + \frac{C}{M} \left(\tilde{h}_{11}^b + \tilde{h}_{22}^b \right) + \tilde{q}^i, \end{aligned} \quad (2.105)$$

$$\begin{aligned} \partial_3 \tilde{p}^f &= -s\rho^f \tilde{v}_3^s - \frac{\eta}{\hat{k}} \left(1 + \frac{\hat{\rho}^E}{\hat{\epsilon}} \hat{\mathcal{L}}^2 \right) \tilde{w}_3 - j\kappa \frac{\hat{\rho}^E}{\hat{\epsilon}} \hat{\mathcal{L}} \tilde{H}_{norm}^V \\ &- \frac{\hat{\rho}^E}{\hat{\epsilon}} \hat{\mathcal{L}} \tilde{J}_3^e + \tilde{f}_3^f, \end{aligned} \quad (2.106)$$

$$\begin{aligned} \partial_3 \tilde{\tau}_{norm}^{b,V} &= j\kappa \frac{S}{K_c} \tilde{\tau}_{33}^b + j\kappa \frac{\rho^f}{\hat{\rho}^E} \tilde{p}^f - j\kappa \frac{C}{M} \left(1 - \frac{S}{K_c} \right) \tilde{p}^f \\ &+ \kappa^2 \frac{1}{s} \frac{4G^{fr} (S + G^{fr})}{K_c} \tilde{v}_{norm}^{s,V} \\ &+ \left(s\rho^b - s^2 (\rho^f)^2 \frac{\hat{k}}{\eta} \right) \tilde{v}_{norm}^{s,V} + s\rho^f \hat{\mathcal{L}} \tilde{E}_{norm}^V \\ &+ \frac{\rho^f}{\hat{\rho}^E} \frac{k_\alpha}{\kappa} \tilde{f}_\alpha^f - \frac{k_\alpha}{\kappa} \tilde{f}_\alpha^b + \frac{j\kappa S^2}{s K_c} \left(\tilde{h}_{11}^b + \tilde{h}_{22}^b \right) \\ &+ \frac{2G^{fr} k_1 k_2}{sj\kappa} \left(\tilde{h}_{21}^b + \tilde{h}_{12}^b \right) \\ &+ \frac{k_1^2}{sj\kappa} \left(K_c \tilde{h}_{11}^b + S \tilde{h}_{22}^b \right) + \frac{k_2^2}{sj\kappa} \left(S \tilde{h}_{11}^b + K_c \tilde{h}_{22}^b \right), \end{aligned} \quad (2.107)$$

$$\partial_3 \tilde{\tau}_{33}^b = j\kappa \tilde{\tau}_{norm}^{b,V} + s\rho^b \tilde{v}_3^s + s\rho^f \tilde{w}_3 - \tilde{f}_3^b, \quad (2.108)$$

$$\begin{aligned} \partial_3 \tilde{E}_{norm}^V &= -s\mu_0 \tilde{H}_{norm}^V - \frac{\kappa^2}{s\hat{\epsilon}} \tilde{H}_{norm}^V + j\kappa \frac{\hat{\rho}^E}{\hat{\epsilon}} \hat{\mathcal{L}} \tilde{w}_3 + \frac{j\kappa}{s\hat{\epsilon}} \tilde{J}_3^e \\ &+ \frac{k_2}{\kappa} \tilde{J}_1^m - \frac{k_1}{\kappa} \tilde{J}_2^m, \end{aligned} \quad (2.109)$$

$$\begin{aligned} \partial_3 \tilde{H}_{norm}^V &= -s\hat{\epsilon} \tilde{E}_{norm}^V + s\rho^f \hat{\mathcal{L}} \tilde{v}_{norm}^{s,V} - \frac{\eta}{\hat{k}} \hat{\mathcal{L}}^2 \tilde{E}_{norm}^V - j\kappa \hat{\mathcal{L}} \tilde{p}^f \\ &- \hat{\mathcal{L}} \frac{k_\alpha}{\kappa} \tilde{f}_\alpha^f - \frac{k_\alpha}{\kappa} \tilde{J}_\alpha^e. \end{aligned} \quad (2.110)$$

We can now capture these two sets of governing seismo-electromagnetic equations (independently for each propagation mode) in the following matrix-vector representation of the two-way wave equation (Wapenaar et al., 2008):

$$\begin{aligned} \partial_3 \tilde{\mathbf{q}}^{H,V}(\mathbf{k}_H, x_3, \omega) &= \tilde{\mathbf{A}}^{H,V}(\mathbf{k}_H, x_3, \omega) \tilde{\mathbf{q}}^{H,V}(\mathbf{k}_H, x_3, \omega) \\ &+ \tilde{\mathbf{d}}^{H,V}(\mathbf{k}_H, x_3, \omega) \delta(x_3 - x_3^s). \end{aligned} \quad (2.111)$$

This two-way wave equation can be used for a variety of waves and fields, and as now demonstrated also for coupled seismo-electromagnetic systems. Equation (2.111) expresses, in matrix-vector notation indicated by the boldface symbols, the vertical variations of the field quantities in $\tilde{\mathbf{q}}^{H,V}(\mathbf{k}_H, x_3, \omega)$, in terms of the medium parameters and radial wavenumbers in system matrix $\tilde{\mathbf{A}}^{H,V}(\mathbf{k}_H, x_3, \omega)$ acting on these field quantities (Woodhouse, 1974; Kennett, 1983; Ursin, 1983; Wapenaar & Grimbergen, 1996). Furthermore, $\tilde{\mathbf{d}}^{H,V}(\mathbf{k}_H, x_3, \omega)$ represents the source vector containing all source terms of the seismo-electromagnetic SH-TE and P-SV-TM modes. The field quantities for each of these modes in $\tilde{\mathbf{q}}^{H,V}$ are continuous across horizontal interfaces. We desire to arrange the field quantities in $\tilde{\mathbf{q}}^{H,V}$ in such a way that the corresponding system matrices $\tilde{\mathbf{A}}^{H,V}$ have optimal symmetry properties. This allows for essential simplifications of our eigenvector analysis, which plays a central role in analytically based layered-Earth modeling. The details of the eigenvector analysis and different eigenvector sets are presented in Chapter 1, section 1.3.

2.2.4.3. Capturing the SH-TE Propagation Mode

We use the following arrangement of the field vector for the SH-TE propagation mode:

$$\tilde{\mathbf{q}}^H = \begin{pmatrix} \tilde{\mathbf{q}}_1^H \\ \tilde{\mathbf{q}}_2^H \end{pmatrix} = \begin{pmatrix} \tilde{\mathbf{v}}^{s,H} \\ \tilde{\mathbf{E}}^H \\ \tilde{\boldsymbol{\tau}}^{b,H} \\ -\tilde{\mathbf{H}}^H \end{pmatrix}, \quad (2.112)$$

where the superscript H indicates that that respective field quantity is an SH-TE mode quantity.

The corresponding system matrix reads

$$\tilde{\mathbf{A}}^H = \begin{pmatrix} \mathbf{O} & \tilde{\mathbf{A}}_{12} \\ \tilde{\mathbf{A}}_{21} & \mathbf{O} \end{pmatrix}, \quad (2.113)$$

which consists of antidiagonal symmetric block matrix operators, reading

$$\tilde{\mathbf{A}}_{12}^H = \begin{pmatrix} \beta & 0 \\ 0 & -\zeta \end{pmatrix}, \quad (2.114)$$

$$\tilde{\mathbf{A}}_{21}^H = \begin{pmatrix} \alpha + \frac{\kappa^2}{\beta} & \chi \\ \chi & -\left(\hat{\eta}^E + \frac{\kappa^2}{\zeta}\right) \end{pmatrix}. \quad (2.115)$$

In equation (2.113), $\mathbf{O}$ is a two-by-two zero-matrix. Furthermore, the following relations hold between the different variables:

$$\alpha = s\rho^c, \quad (2.116)$$

$$\hat{\rho}^c = \rho^b - \frac{(\rho^f)^2}{\rho^E}, \quad (2.117)$$

$$\rho^E = \frac{\eta}{s\hat{k}}, \quad (2.118)$$

$$\chi = s\rho^f \hat{\mathcal{L}}, \quad (2.119)$$

$$\beta = \frac{s}{G^{fr}}, \quad (2.120)$$

$$\zeta = \hat{\sigma}^m + s\mu, \quad (2.121)$$

$$\hat{\eta}^E = \hat{\sigma}^e + s\varepsilon, \quad (2.122)$$

$$\hat{\varepsilon} = \varepsilon_0 \varepsilon_r + \frac{\hat{\sigma}^e}{s} - \rho^E \hat{\mathcal{L}}^2 = \varepsilon + \frac{\hat{\sigma}^e}{s} - \rho^E \hat{\mathcal{L}}^2 = \frac{\hat{\eta}^E}{s} - \rho^E \hat{\mathcal{L}}^2. \quad (2.123)$$

The SH-TE source vector in the horizontal wavenumber-frequency domain then reads

$$\tilde{\mathbf{d}}^H = \left(\tilde{d}_1^H, \tilde{d}_2^H, \tilde{d}_3^H, \tilde{d}_4^H \right)^t, \quad (2.124)$$

where

$$\tilde{d}_1^H = -jk_1 \left[\tilde{h}_{23}^b + \tilde{h}_{31}^b \right] + jk_2 \left[\tilde{h}_{13}^b + \tilde{h}_{31}^b \right], \quad (2.125)$$

$$\tilde{d}_2^H = -jk_\alpha \tilde{\mathcal{J}}_\alpha^m, \quad (2.126)$$

$$\begin{aligned} \tilde{d}_3^H &= jk_1 \left[\tilde{f}_2^b - \frac{\rho^f}{\rho^E} \tilde{f}_2^f \right] - jk_2 \left[\tilde{f}_1^b - \frac{\rho^f}{\rho^E} \tilde{f}_1^f \right] \\ &+ \frac{1}{s} \left[-k_1^2 + k_2^2 \right] \left[G^{fr} \tilde{h}_{21}^b + G^{fr} \tilde{h}_{12}^b \right] \\ &- \frac{k_1 k_2}{s} \left[-2G^{fr} \tilde{h}_{11}^b + 2G^{fr} \tilde{h}_{22}^b \right], \end{aligned} \quad (2.127)$$

$$\tilde{d}_4^H = -\hat{\mathcal{L}} \left[-jk_1 \tilde{f}_2^f + jk_2 \tilde{f}_1^f \right] + jk_1 \tilde{\mathcal{J}}_2^e - jk_2 \tilde{\mathcal{J}}_1^e - \frac{1}{s\mu_0} \kappa^2 \tilde{\mathcal{J}}_3^m. \quad (2.128)$$

2.2.4.4. Capturing the P-SV-TM Propagation Mode

For the P-SV-TM propagation mode, we use the following arrangement of the field vector:

$$\tilde{\mathbf{q}}^V = \begin{pmatrix} \tilde{\mathbf{q}}_1^V \\ \tilde{\mathbf{q}}_2^V \end{pmatrix} = \begin{pmatrix} \tilde{\mathbf{v}}_3^s \\ \tilde{\mathbf{w}}_3 \\ \tilde{\boldsymbol{\tau}}_{norm}^{b,V} \\ \tilde{\mathbf{H}}_{norm}^V \\ \tilde{\boldsymbol{\tau}}_{33}^b \\ -\tilde{\mathbf{p}}^f \\ \tilde{\mathbf{v}}_{norm}^{s,V} \\ \tilde{\mathbf{E}}_{norm}^V \end{pmatrix}, \quad (2.129)$$

where the superscript V indicates that these field quantities are P-SV-TM mode quantities.

These choices result in the following P-SV-TM system matrix:

$$\tilde{\mathbf{A}}^V = \begin{pmatrix} \mathbf{O} & \tilde{\mathbf{A}}_{12}^V \\ \tilde{\mathbf{A}}_{21}^V & \mathbf{O} \end{pmatrix}, \quad (2.130)$$

where the submatrices are defined as

$$\tilde{\mathbf{A}}_{12}^V = \begin{pmatrix} \frac{s}{K_c} & -s \frac{C}{MK_c} & \frac{j\kappa S}{K_c} & 0 \\ -s \frac{C}{MK_c} & \frac{s}{M} \left[1 + \frac{C^2}{MK_c} \right] + \frac{\hat{k}}{\eta} \kappa^2 & -j\kappa \frac{\rho^f}{\hat{\rho}^E} + j\kappa \frac{C}{M} \left[1 - \frac{S}{K_c} \right] & j\kappa \hat{\mathcal{L}} \\ \frac{j\kappa S}{K_c} & -j\kappa \frac{\rho^f}{\hat{\rho}^E} + j\kappa \frac{C}{M} \left[1 - \frac{S}{K_c} \right] & \frac{1}{s} \frac{4G_{fr}(S + G_{fr})}{K_c} \kappa^2 + s\hat{\rho}^c & \chi \\ 0 & j\kappa \hat{\mathcal{L}} & \chi & -\hat{\eta}^E \end{pmatrix}, \quad (2.131)$$

and

$$\tilde{\mathbf{A}}_{21}^V = \begin{pmatrix} s\rho^b & s\rho^f & j\kappa & 0 \\ s\rho^f & \frac{s\rho^E \hat{\eta}^E}{\hat{\zeta}} & 0 & j\kappa \frac{\hat{\xi}}{\hat{\zeta}} \\ j\kappa & 0 & \beta & 0 \\ 0 & j\kappa \frac{\hat{\xi}}{\hat{\zeta}} & 0 & -\zeta - \frac{1}{\hat{\zeta}} \kappa^2 \end{pmatrix}. \quad (2.132)$$

The following additional relations hold between different variables:

$$\xi = \frac{\eta}{\hat{k}} \hat{\mathcal{L}} = s\rho^E \hat{\mathcal{L}}, \quad (2.133)$$

$$\varsigma = \hat{\eta}^E - \xi \hat{\mathcal{L}}, \quad (2.134)$$

$$\frac{\hat{\eta}^e}{\hat{\zeta}} = 1 + \frac{\xi \hat{\mathcal{L}}}{\hat{\zeta}}. \quad (2.135)$$

The corresponding P-SV-TM source vector in the horizontal wavenumber-frequency domain is given by

$$\tilde{\mathbf{d}}^V = (\tilde{d}_1^V, \tilde{d}_2^V, \tilde{d}_3^V, \tilde{d}_4^V, \tilde{d}_5^V, \tilde{d}_6^V, \tilde{d}_7^V, \tilde{d}_8^V)^T, \quad (2.136)$$

where

$$\tilde{d}_1^V = \frac{S}{(S + 2G^{fr})} [\tilde{h}_{11}^b + \tilde{h}_{22}^b] + \tilde{h}_{33}^b, \quad (2.137)$$

$$\tilde{d}_2^V = \frac{j}{s\rho^E} k_a \tilde{f}_a^f + \frac{C}{M} [\tilde{h}_{11}^b + \tilde{h}_{22}^b] \left[1 - \frac{S}{(S + 2G^{fr})} \right] + \tilde{q}^i, \quad (2.138)$$

$$\begin{aligned} \tilde{d}_3^V &= \frac{1}{\kappa} \frac{\rho^f}{\hat{\rho}^E} k_a \tilde{f}_a^f - \frac{1}{\kappa} k_a \tilde{f}_a^b - \frac{2j}{s} \frac{k_1 k_2}{\kappa} [G^{fr} \tilde{h}_{21}^b + G^{fr} \tilde{h}_{12}^b] \\ &\quad - \frac{j\kappa}{s} \frac{2G^{fr} S}{(S + 2G^{fr})} [\tilde{h}_{11}^b + \tilde{h}_{22}^b] - \frac{2jG^{fr}}{s\kappa} [k_1^2 \tilde{h}_{11}^b + k_2^2 \tilde{h}_{22}^b], \end{aligned} \quad (2.139)$$

$$\tilde{d}_4^V = -\frac{1}{\kappa} [\hat{\mathcal{L}} k_a \tilde{f}_a^f + k_a \tilde{J}_a^e] \quad (2.140)$$

$$\tilde{d}_5^V = -\tilde{f}_3^b, \quad (2.141)$$

$$\tilde{d}_6^V = \frac{\hat{\rho}^E}{\hat{\epsilon}} \hat{\mathcal{L}} \tilde{J}_3^e - \tilde{f}_3^f, \quad (2.142)$$

$$\tilde{d}_7^V = \frac{k_a}{\kappa} [\tilde{h}_{a3}^b + \tilde{h}_{3a}^b], \quad (2.143)$$

$$\tilde{d}_8^V = \frac{j\kappa}{s\hat{\epsilon}} \tilde{J}_3^e + \frac{1}{\kappa} [k_2 \tilde{J}_1^m - k_1 \tilde{J}_2^m]. \quad (2.144)$$

These SH-TE and P-SV-TM field vectors, system matrices, and source vectors are key ingredients for the eigenvalue decomposition and eigenvector analysis that form the foundation for many analytically based layered-Earth numerical modeling codes (Haartsen & Pride, 1997; Grobbe et al., 2016a). The system matrices $\tilde{\mathbf{A}}^{H,V}$ can be decomposed into their eigenvector and eigenvalue matrices as $\tilde{\mathbf{A}}^{H,V} = \tilde{\mathbf{L}}^{H,V} \tilde{\mathbf{\Lambda}}_1^{H,V} \{\tilde{\mathbf{L}}^{H,V}\}^{-1}$, where $\tilde{\mathbf{L}}^{H,V}$ and $\{\tilde{\mathbf{L}}^{H,V}\}^{-1}$ represent the composition and decomposition matrices. These matrices are composed of the eigenvectors of the SH-TE and P-SV-TM modes, and form the foundation for wavefield decomposition into one-way wavefields (see, e.g., Grobbe et al., 2016b). $\tilde{\mathbf{\Lambda}}_1^{H,V}$ represents a diagonal matrix containing the eigenvalues of the system:

$$\tilde{\mathbf{\Lambda}}_1^{H,V} = \begin{pmatrix} -\tilde{\mathbf{\Lambda}}^{H,V} & \mathbf{O} \\ \mathbf{O} & \tilde{\mathbf{\Lambda}}^{H,V} \end{pmatrix}, \quad (2.145)$$

with

$$\tilde{\mathbf{\Lambda}}^H = \begin{pmatrix} \Gamma_{SH} & 0 \\ 0 & \Gamma_{TE} \end{pmatrix} \quad (2.146)$$

for the SH-TE mode, and

$$\tilde{\mathbf{A}}^V = \begin{pmatrix} \Gamma_{Pf} & 0 & 0 & 0 \\ 0 & \Gamma_{Ps} & 0 & 0 \\ 0 & 0 & \Gamma_{SV} & 0 \\ 0 & 0 & 0 & \Gamma_{TM} \end{pmatrix} \quad (2.147)$$

for the P-SV-TM mode. Γ_w represent the vertical wave-numbers, which can be related to the spherical wavenumbers γ_w for a certain type of field w via

$$\Gamma_w^2 = \gamma_w^2 + \kappa^2 \quad (2.148)$$

$$\gamma_w^2 = \Gamma_w^2 - \kappa^2, \quad (2.149)$$

where

$$\gamma_w = \frac{s}{\hat{c}_w} \quad (2.150)$$

and where w can either represent the horizontally polarized shear wave SH , the transverse electric mode TE , the fast compressional wave P_f , the Biot slow compressional wave P_s , the vertically polarized shear wave SV , or the transverse magnetic mode TM . $\hat{c}_w$ denotes the complex wave-field velocity for the specific field type w . Pride and Haartsen (1996) have determined the complex velocities for each field type, which we can rewrite for our field quantities and our definition of the Fourier transform as

$$\hat{c}_{SH} = \sqrt{\frac{2}{\frac{\hat{\rho}^c}{G^{fr}} + \frac{\zeta\eta^E}{s^2} + \sqrt{\left[\frac{\hat{\rho}^c}{G^{fr}} - \frac{\zeta\eta^E}{s^2}\right]^2 - \frac{4\zeta\chi^2}{s^3 G^{fr}}}}} \quad (2.151)$$

$$\hat{c}_{TE} = \sqrt{\frac{2}{\frac{\hat{\rho}^c}{G^{fr}} + \frac{\zeta\eta^E}{s^2} + \sqrt{\left[\frac{\hat{\rho}^c}{G^{fr}} - \frac{\zeta\eta^E}{s^2}\right]^2 - \frac{4\zeta\chi^2}{s^3 G^{fr}}}}} \quad (2.152)$$

$$\hat{c}_{Pf} = \sqrt{\frac{2(HM - C^2)}{\hat{\nu} - \sqrt{\hat{\nu}^2 - 4(HM - C^2)\left[\hat{\rho}^b \frac{\hat{\rho}^E \eta^E}{s\hat{e}} - (\rho^f)^2\right]}}} \quad (2.153)$$

$$\hat{c}_{Ps} = \sqrt{\frac{2(HM - C^2)}{\hat{\nu} + \sqrt{\hat{\nu}^2 - 4(HM - C^2)\left[\hat{\rho}^b \frac{\hat{\rho}^E \eta^E}{s\hat{e}} - (\rho^f)^2\right]}}} \quad (2.154)$$

$$\hat{c}_{SV} = \hat{c}_{SH} \quad (2.155)$$

$$\hat{c}_{TM} = \hat{c}_{TE}, \quad (2.156)$$

where $\hat{\nu} = \rho^b M + \frac{\hat{\rho}^E \eta^E H}{\hat{\zeta}} - 2\rho^f C$. Due to the fact that, in

the layered-Earth system discussed here, we are dealing with an isotropic and laterally invariant medium, the horizontally and vertically polarized shear wave velocities are equal to each other. The same holds for the transverse electric and transverse magnetic (diffusive) “velocities.” Haartsen and Pride (1997) published the first set of eigenvectors for layered-Earth seismoelectric and electroseismic modeling. Grobbe et al. (2016a) have power-flux normalized these eigenvector sets, and have also derived a novel, alternative set of eigenvectors. The power-flux normalization was proven to result in more stable and accurate numerical results. Furthermore, Grobbe et al. (2016a) demonstrate that their novel-derived eigenvector sets capture the physics of the seismoelectric phenomenon correctly, even when the seismoelectric coupling coefficient is equal to zero. In this case, there is no coupling between mechanical and electromagnetic fields, and the Grobbe et al. (2016a) eigenvector sets model the purely poroelastic and purely electromagnetic systems correctly, as opposed to models based on the Haartsen and Pride (1997) eigenvector sets. Additionally, Grobbe et al. (2016a) show that in scenarios where there is little coupling between mechanical and electromagnetic fields (e.g., low porosity values or high electrolyte concentrations), the novel eigenvectors of Grobbe et al. (2016a) model all events correctly and clean, whereas the Haartsen and Pride (1997) eigenvector sets result in numerical noise levels that are of similar amplitude as the weak events in the data, making a distinction between noise and physically weak events impossible. Given the generally weak seismoelectric converted fields, as well as to provide optimal insight into the physics of seismoelectric and electroseismic phenomena, it is desirable to use a modeling scheme that is robust and accurate under all scenarios. We therefore strongly recommend using the novel eigenvector sets of Grobbe et al. (2016a) for future seismoelectric and electroseismic layered-Earth modeling efforts, as opposed to the eigenvector sets of Haartsen and Pride (1997). The reader is referred to the aforementioned references for the complete expressions of these eigenvector sets.

For illustrative purposes, we briefly present a numerical modeling result that is based on the above presented theoretical framework. This is just a tip of the iceberg of the endless seismoelectric numerical modeling possibilities. Other seismoelectric numerical modeling examples (both analytically based and numerically based) can be found in Section 3 of this book. Based on the above theory, and a global reflection algorithm, we have developed the layered-Earth, analytically based numerical modeling code ESSEMOD (Grobbe, 2016; Grobbe et al., 2016a). The code is also capable of modeling seismoelectric

wavefields based on the eigenvector sets of Haartsen and Pride (1997), a power-flux normalized version of these eigenvectors (Grobbe et al., 2016a), and the alternative, newly derived eigenvector set mentioned above (Grobbe et al., 2016a).

Figure 2.1 displays the results of a numerical reciprocity test using ESSEMOD. The model consists of a single layer over a halfspace, including a free surface. The interface between the layer and the halfspace is located at $z = 1200$ m. Figure 2.1a displays the electric field in the x_1 -direction due to a seismic bulk force source in the same direction: E_1^{f1b} . The source is located at a depth of 100 m,

and the receivers at a depth of 700 m. Figure 2.1b shows the modeling results for the reciprocal configuration and source-receiver combination, the source at 700 m depth and the receivers located at 100 m depth. This time, we observe the results of the particle velocity in the x_1 -direction due to an electrical current source in the same direction: ν_1^{j1e} . Figure 2.1c displays a zero-offset trace overlay, where we can clearly observe that the E_1^{f1b} -response (red-solid) and the ν_1^{j1e} -response perfectly match. This partially validates the consistency and correctness of the modeling code ESSEMOD. More validation results can be found in Grobbe (2016).

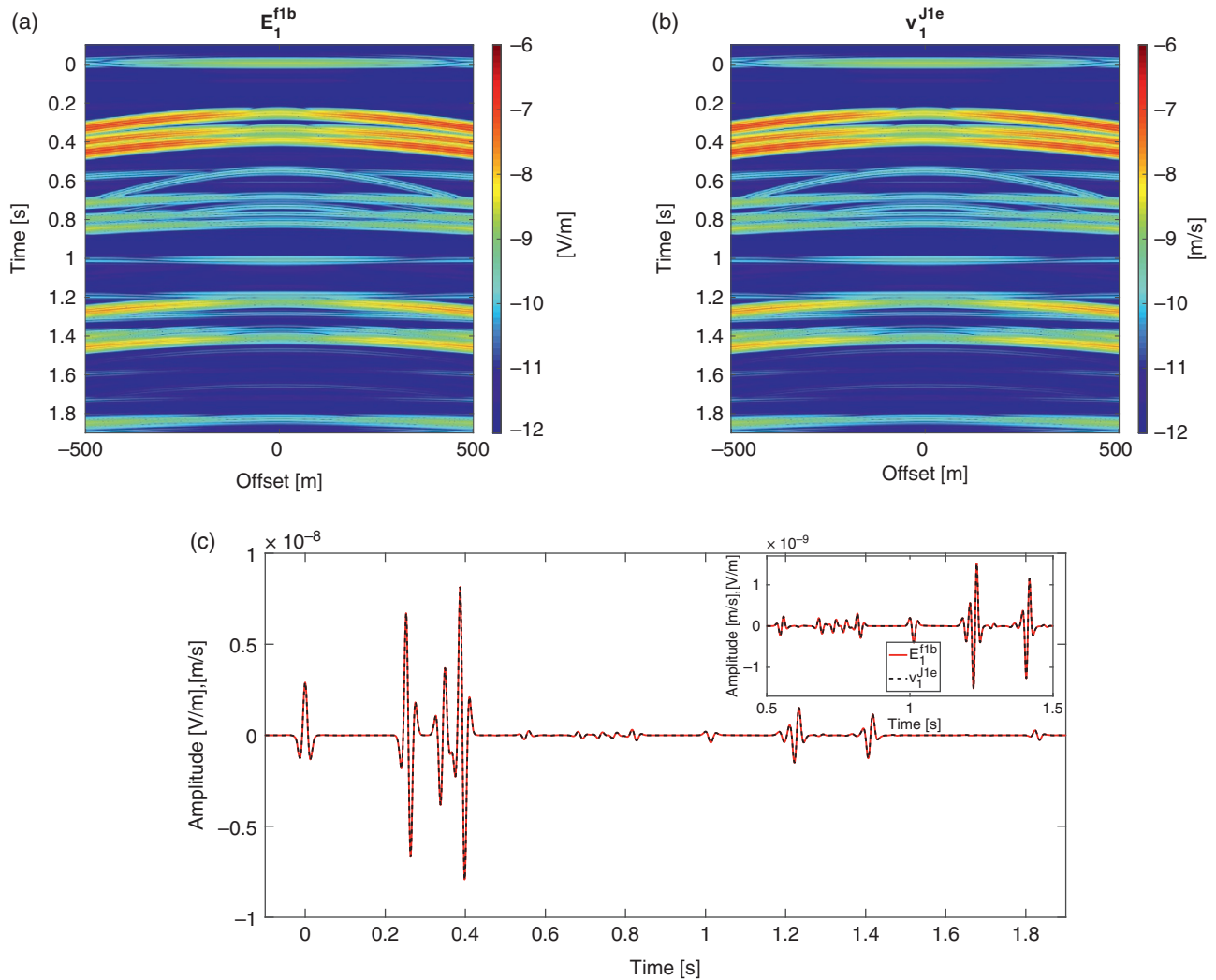


Figure 2.1 Seismoelectric numerical reciprocity tests using ESSEMOD, for a model containing a single layer over a halfspace (interface at $z = 1,200$ m), including a free surface: (a) E_1^{f1b} source-receiver combination, with the sources located at 100 m and receivers at 700 m depth; (b) reciprocal ν_1^{j1e} with the sources located at 700 m depth and the receivers at 100 m depth; (c) zero-offset trace overlay displaying identical reciprocal modeling results. Red-solid: E_1^{f1b} , Black-dashed: ν_1^{j1e} .

2.2.5. Poroacoustic Theory and Partial Saturation

In this subsection, we introduce one way to simplify the seismoelectric modeling by computing the electrical field associated with the propagation of an acoustic wave. Furthermore, we will discuss a variety of ways to extend the theory to partially saturated conditions (see also Revil et al., 2015). To derive the simplified poroacoustic modeling framework, we first need to express the propagation of a seismic wave using an acoustic approximation. We write $P(\mathbf{r}, t)$ (in Pa) as the pressure perturbation (true pressure minus the equilibrium pressure), where $\mathbf{r}$ denotes the spatial vector, and t is time. The volumetric strain θ is related to the displacement $\mathbf{u}(\mathbf{r}, t)$ by

$$\theta = \frac{\Delta V}{V} = \nabla \cdot \mathbf{u}, \quad (2.157)$$

where $\frac{\Delta V}{V}$ represents a relative variation of volume of the porous material during the passage of the seismic wave. The compressibility of the material is defined, in isothermal conditions, as

$$\frac{1}{K_u} = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{T, m_f}, \quad (2.158)$$

where m_f denotes the mass of fluid per unit volume of material, and K_u denotes the undrained bulk modulus (in Pa). Therefore, we compute the effect of the passage of a seismic wave on fluid flow in two steps: (1) An instantaneous deformation, in undrained conditions, during the passage of the seismic wave and (2) a slow deformation corresponding to the flow of the fluid in the pore space. The confining pressure is related to the stress tensor $\mathbf{T}$ by

$$P = -\frac{1}{3} \text{Trace } \mathbf{T}. \quad (2.159)$$

The undrained compression bulk modulus K_u (in Pa) is identical to the earlier defined K_G , and can alternatively be defined by

$$K_u = \frac{K^f(K^s - K) + \phi K(K^s - K^f)}{K^f(1 - \phi - K/K^s) + \phi K^s}, \quad (2.160)$$

where K denotes the bulk modulus of the skeleton (drained bulk modulus; in Pa), K^s and K^f denote the bulk moduli of the solid and fluid phases, respectively, and ϕ denotes the connected porosity. The assumptions corresponding to this deformation of the porous material during its undrained deformation are those of the Gassmann theory (Gassmann, 1951): (1) The porous material is macroscopically homogeneous, monomineralic, and isotropic; (2) the pore space is interconnected; and (3) the pore space is filled with a frictionless fluid. We use Hooke's law for the stress tensor,

$$\mathbf{T} = 3K_u \text{vol}(\boldsymbol{\epsilon}) + 2G^{fr} \text{dev}(\boldsymbol{\epsilon}), \quad (2.161)$$

where $\text{vol}(\boldsymbol{\epsilon})$ and $\text{dev}(\boldsymbol{\epsilon})$ denote the volumetric strain tensor and the deviatoric (traceless) strain tensor, respectively, and G^{fr} describes the shear modulus of the skeleton (frame) of the porous material. The macroscopic pressure perturbation is related to the displacement by

$$P = -K_u \nabla \cdot \mathbf{u}. \quad (2.162)$$

However, to be compatible with the P-wave velocity, this equation needs to be replaced by,

$$P = -\left(K_u + \frac{4}{3}G^{fr}\right) \nabla \cdot \mathbf{u}. \quad (2.163)$$

Equation (2.161) corresponds to Hooke's law in the acoustic approximation and is valid for small deformation ($\theta \ll 1$). To get the field equation for the pressure perturbation P , we need to combine this equation with Newton's law, which is given by

$$-\nabla P + \mathbf{F} = \rho^b \mathbf{u}, \quad (2.164)$$

where $\mathbf{F}$ is a force source vector and $\mathbf{u}$ corresponds to the acceleration of the seismic wave. As we need to express the divergence of the displacement in terms of pressure, we want to take the divergence of equation (2.164). Subsequently substituting equation (2.163) and reorganizing results in

$$-\nabla \cdot \left(\frac{1}{\rho^b} \nabla P \right) + \nabla \cdot \left(\frac{1}{\rho^b} \mathbf{F} \right) = \frac{\partial^2}{\partial t^2} (\nabla \cdot \mathbf{u}), \quad (2.165)$$

$$-\nabla \cdot \left(\frac{1}{\rho^b} \nabla P \right) + \nabla \cdot \left(\frac{1}{\rho^b} \mathbf{F} \right) = -\frac{1}{K_u + \frac{4}{3}G^{fr}} \left(\frac{\partial^2 P}{\partial t^2} \right), \quad (2.166)$$

$$\begin{aligned} & \left(\frac{\partial^2 P}{\partial t^2} \right) - \left(K_u + \frac{4}{3}G^{fr} \right) \nabla \cdot \left(\frac{1}{\rho^b} \nabla P \right) \\ & = -\left(K_u + \frac{4}{3}G^{fr} \right) \nabla \cdot \left(\frac{1}{\rho^b} \mathbf{F} \right), \end{aligned} \quad (2.167)$$

which yields the acoustic wave equation with inhomogeneous source term $f(\mathbf{r}, t)$, that specifies the location and time history of the source,

$$\left(\frac{\partial^2 P}{\partial t^2} \right) - \left(K_u + \frac{4}{3}G^{fr} \right) \nabla \cdot \left(\frac{1}{\rho^b} \nabla P \right) = f(\mathbf{r}, t), \quad (2.168)$$

$$f(\mathbf{r}, t) = -\left(K_u + \frac{4}{3}G^{fr} \right) \nabla \cdot \left(\frac{1}{\rho^b} \mathbf{F} \right). \quad (2.169)$$

If we wish to determine the displacement of the fluid, we can for instance consider a wave with a sinusoidal time dependence given by

$$P(\mathbf{r}, t) = \hat{P}(\mathbf{r}, \omega) \exp(j\omega t). \quad (2.170)$$

Inserting equation (2.170) into equation (2.168), neglecting the spatial variation in the density and the source term, and expressing the second-order temporal derivative in the frequency-domain shows that the amplitudes of the seismic waves obey the scalar Helmholtz equation,

$$\nabla^2 \hat{P}(\mathbf{r}, \omega) + \frac{\omega^2}{c_p^2} \hat{P}(\mathbf{r}, \omega) = 0, \quad (2.171)$$

where

$$c_p = \sqrt{\frac{K_u + \frac{4}{3}G^r}{\rho^b}}. \quad (2.172)$$

If we consider Newton's law, equation (2.168), we obtain the following equation for the particle displacement:

$$\mathbf{u}(\mathbf{r}, t) = \frac{\rho^b}{\omega^2} \nabla \hat{P}(\mathbf{r}, \omega) \exp(j\omega t). \quad (2.173)$$

Because the curl of a gradient is always equal to zero, we have the property $\nabla \times \mathbf{u}(\mathbf{r}, t) = 0$. The displacement is irrotational, which means that the pressure wave is purely longitudinal and corresponds to a P-wave with a seismic velocity given by equation (2.172). We need now to express the pressure change on the material to a pore fluid pressure change. In the undrained regime of poroelasticity, the pressure P is related to the so-called undrained pore fluid pressure p by (Wang, 2000),

$$p = BP, \quad (2.174)$$

where $0 \leq B \leq 1$ is called the Skempton coefficient given by

$$B = \frac{1 - K/K_u}{1 - K/K^s}. \quad (2.175)$$

The passage of the wave generates a confining pressure fluctuation P to a representative elementary volume of the rock. This change in confining pressure generates in turn a change in the pore fluid pressure and therefore the flow of pore water. Darcy's law provides the needed constitutive equation to determine the flux of water through the porous material:

$$\dot{\mathbf{w}} = -\frac{k_s}{\eta} \nabla p, \quad (2.176)$$

where $\mathbf{w}$ denotes the filtration displacement (i.e., the relative displacement times the porosity), $\dot{\mathbf{w}}$ represents the Darcy velocity (flux of water in a Lagrangian framework associated with the deformation of the skeleton), k_s denotes the permeability at saturation, and η is the dynamic viscosity of the pore fluid.

The flow of the pore water drags the excess of charge contained in the pore water, generating a streaming current density (we will neglect the electro-osmotic effect in the coupling equation since the only source of the electrical field is the one produced by the seismic wave, which is expected to be very small). We consider the occurrence of the associated electrical disturbances in the quasi-static limit of the Maxwell equations as commonly done in self-potential studies. The following equation holds for the quasi-static approximation (induction terms neglected):

$$\nabla \cdot (\sigma \nabla \varphi) = \nabla \cdot \mathbf{j}_s, \quad (2.177)$$

where the electrical field is given by $\mathbf{E} = -\nabla \varphi$, and σ (in S m^{-1}) denotes the electrical conductivity of the porous medium. The dynamic streaming current density of electrokinetic nature $\mathbf{j}_s$ (in A m^{-2}), associated with the flow of the pore water, is given by

$$\mathbf{j}_s = \hat{Q}_V \dot{\mathbf{w}} = -\frac{\hat{Q}_V k_s}{\eta} \nabla p, \quad (2.178)$$

where $\hat{Q}_V$ denotes the effective charge per unit pore volume that is dragged by the flow of pore water relative to the solid phase. This parameter can be related to the permeability by $\log_{10} \hat{Q}_V = -9.2349 - 0.8219 \log_{10} k_0$ (Revil & Mahardika, 2013). We neglect inertial terms, which is a reasonable approximation when dealing with frequencies below 5 kHz (see Revil & Mahardika, 2013) for an example calculation showing the importance of the inertial terms). The electrical potential φ (in V) is governed by the following Poisson equation:

$$\nabla \cdot (\sigma \nabla \varphi) = \nabla \cdot \mathbf{j}_s = -\nabla \cdot \left(\frac{\hat{Q}_V k_s}{\eta} \nabla (BP) \right). \quad (2.179)$$

Solving equation (2.179) is required to understand the electrical disturbance associated with the passage of an acoustic seismic wave through a heterogeneity such as an interface. In this case, the corresponding electrical field is called the seismoelectric conversion (or interface response field).

At saturation, the electrical conductivity of the porous material is given by

$$\sigma = \frac{1}{F} \sigma^f + \left(\frac{1}{F\phi} \right) \rho^s B \text{CEC}, \quad (2.180)$$

where B (in $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$) represents the apparent mobility for the counterions, F is the electrical formation factor (a power-law function of porosity), and CEC denotes the cation exchange capacity of the material (in C kg^{-1}).

We consider an unsaturated material with two immiscible fluid phases: water that is considered the wetting phase for the solid and a nonwetting and insulating fluid phase like air or oil. The density of the pore fluid ρ^f depends on

saturation according to $\rho^f = (1 - s_w)\rho^g + s_w\rho^w$, where s_w denotes the saturation of the water (wetting) phase, and ρ^g and ρ^w represent the mass density of the gas or oil (non-wetting) and water (wetting) phase, respectively.

The undrained bulk modulus K_u , the bulk modulus of the pore fluid K^f , and the dynamic viscosity of the pore fluid η , are related to the properties of the gas (nonwetting phase; subscript g) and water (wetting phase; subscript w) by the following relationships (Revil & Mahardika, 2013):

$$K_u(s_w) = \frac{K^f(s_w)(K^s - K) + s_w\phi K(K^s - K^f(s_w))}{K^f(s_w)(1 - \phi - K/K^s) + \phi K^s}, \quad (2.181)$$

with

$$K^f(s_w) = \left(\frac{1 - s_w}{K^g} + \frac{s_w}{K^w} \right), \quad (2.182)$$

$$\eta(s_w) = \eta^g \left(\frac{\eta^w}{\eta^g} \right)^{s_w}, \quad (2.183)$$

where equation (2.182) corresponds to Wood's formula (Wood, 1930) and equation (2.187) can be found in Teja and Rice (1981). The saturation dependence of the Skempton coefficient depends on the saturation dependence of the undrained bulk modulus K_u . We consider that the capillary pressure is at equilibrium (the deformation of the meniscus under the flow of the fluid phases is neglected). The saturation can be related to the capillary pressure by the Brook and Corey relationship,

$$s_w = \begin{cases} s_r + (1 - s_r) \left(\frac{p_c}{p_e} \right)^\lambda, & p_c \geq p_e, \\ 1, & p_c < p_e, \end{cases} \quad (2.184)$$

or alternatively $p_c = p_e s_e^{-1/\lambda}$ (or $s_e = (p_c/p_e)^{-\lambda}$, $p_c \geq p_e$ (see equation 12 of Brooks & Corey, 1964), where $s_e = (s_w - s_r)/(1 - s_r)$ denotes the effective or reduced water saturation, s_r the irreducible water saturation, p_c the capillary pressure, and p_e the capillary entry pressure (i.e., the critical pressure needed to displace the water phase by the gas phase when the porous material is fully water saturated).

Next, we need to determine the effect of saturation on the electrical conductivity. We investigate the following model:

$$\sigma = \frac{1}{F} s_w^n \sigma_w + s_w^{n-1} \sigma_s, \quad (2.185)$$

where n is called the saturation exponent and σ_s denotes the surface conductivity at saturation. The effect of saturation on the excess of electrical charges per unit volume is described by extending the empirical relationship between these two parameters in unsaturated conditions

$$\hat{Q}_V(s_w) = \frac{\hat{Q}_V}{s_w}. \quad (2.186)$$

The permeability can be computed as the product of the permeability at saturation k_s and a relative permeability k_r that depends only on the relative water saturation,

$$k(s_w) = k_s k_r(s_w), \quad (2.187)$$

with $k_r(s_w) = s_e^{n+2}$. In the Brooks and Corey (1964) model, the relative permeability is also given by a power law relationship,

$$k_r(s_w) = \left(\frac{p_c}{p_e} \right)^{-2-3\lambda} = s_e^r, \quad (2.188)$$

where

$$r = \frac{2 + 3\lambda}{\lambda}, \quad (2.189)$$

with λ representing the pore size distribution index. We identify the following equalities:

$$\lambda = \frac{2}{n-1}, r = n + 2. \quad (2.190)$$

This result is very important because it provides an explicit relationship between a hydraulic parameter and an electrical parameter. We now turn our attention to the capillary entry pressure. The capillary entry pressure is usually given by

$$p_e = \frac{2\gamma}{r_c}, \quad (2.191)$$

where γ represents the surface tension between water and air, $(71.99 \pm 0.05) \cdot 10^{-3} \text{ N m}^{-1}$, and r_c , the percolation length scale, represents the smallest pore of the set of pores through which water percolates in the porous material. The capillary entry pressure and the permeability can be related to each other by

$$p_e = \frac{2\gamma}{\sqrt{226F}} k_s^{-0.5}. \quad (2.192)$$

We can now apply the previously developed acousto-electric model with partial saturation in a beamforming approach (e.g., Sava & Revil, 2012) to localize a water saturation front. The beamforming technique enables us to focus seismic energy at a desired location and at a known time knowing approximately the velocity model. Through the scanning of various points, this technique enables us to identify whether the point of focus is located close to a saturation front or not. We solve the partial differential equations for the mechanical and electrical problems in the frequency domain. The velocity model does not need to be known perfectly.

We compute the distribution of the model parameters given the porosity, fluid permeability, and saturation distributions, and subsequently solve for the confining pressure $P(r, t)$ of the solid phase and the corresponding pore

fluid pressure $p(r, t)$ during the acoustic wave propagation. Next, we compute the electrical potential by solving the Poisson equation coupled to the poroacoustic problem for the frequency range of 8 to 800 Hz. Finally, we compute the inverse Fourier transform (FFT^{-1}) to obtain the time series of the seismic displacement components u_1 and u_3 , and the time series of the electrical potential response. Figure 2.2 displays the results of a cross-well transmission experiment, without beamforming. Figure 2.2a displays the geometry of the experiment, with a seismic source located downhole in the well on the left, and electrical receivers located in the well on the right. The saturation front indicates the oil/water front in this water-flooding experiment (i.e., water is injected in the well on the left and oil is produced in the well on the right; see also Revil

et al., 2014). Figure 2.2b displays the recorded electrical potential time series, which displays the records of both the coseismic field and the seismoelectric conversion occurring at the interface (around time $t = 58$ ms, the one-way time from seismic source to saturation front). Figure 2.2c displays the fluid pressure field located at the saturation front at the center of the Fresnel zone. This arrival corresponds with the arrival time of the seismoelectric conversion in Figure 2.2b. The sensitivity of seismoelectric signals for identification of contrasts between wetting and nonwetting fluids has also been successfully shown using laboratory experiments, validated by theoretical formulations and modeling (Smeulders et al., 2014). This sensitivity was further demonstrated by numerical analyses of seismoelectric thin-bed responses

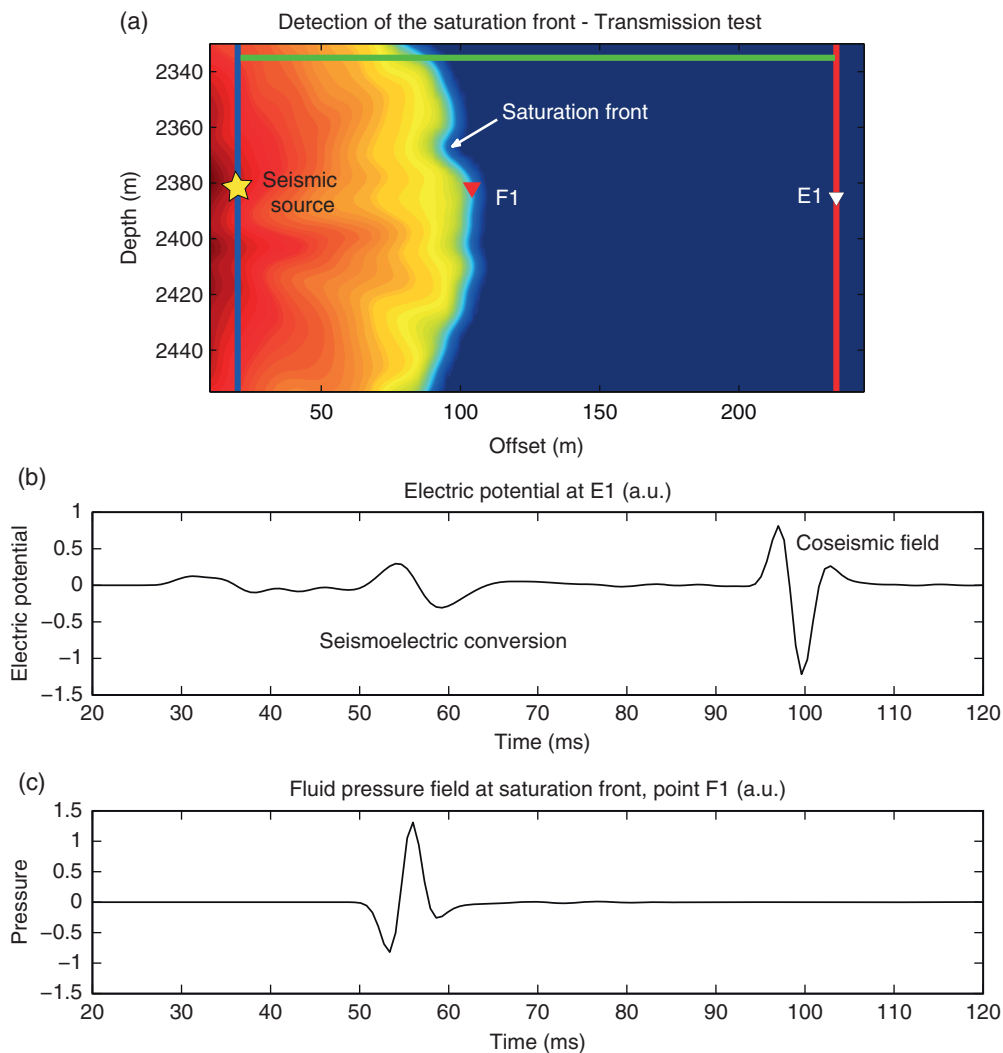


Figure 2.2 Transmission experiment with the seismic source located in Well A and the electrical receiver in Well B: (a) Geometry of the modeling experiment; (b) time series of the electrical potential showing both the coseismic field and the seismoelectric conversion occurring at the saturation front; (c) fluid pressure field at a point located on the saturation front at the center of the Fresnel zone.

and seismoelectric responses from very thin layers of oil over water (Grobbe & Slob, 2016).

Next, we carry out a seismoelectric beamforming experiment for the same model as displayed in Figure 2.2a. Figure 2.3a to e show the results of seismic beamforming using seismic sources located in three wells. Figure 2.3f shows the resulting electrical potential distribution at the so-called focus-time ($t = 0$ s), for which the seismic energy is beamformed. We can observe a strong dipolar field associated with the seismic beamforming.

In Figure 2.4, we repeat this operation for a set of scanning points located at the same depth and crossing the position of the saturation front. Our analysis shows that the strongest seismoelectric conversion occurs at the

position of the saturation front, where a contrast in saturation is located. Therefore, we can conclude that by scanning a reservoir one could determine the position of the saturation front and monitor its progression over time. Obviously, the seismoelectric focusing approach is valuable for other applications as well.

2.3. CONCLUSIONS

The governing equations as presented in this chapter provide a proper framework to describe the seismoelectric coupling phenomenon and wave propagation in arbitrary porous media with varying degrees of saturation. We have

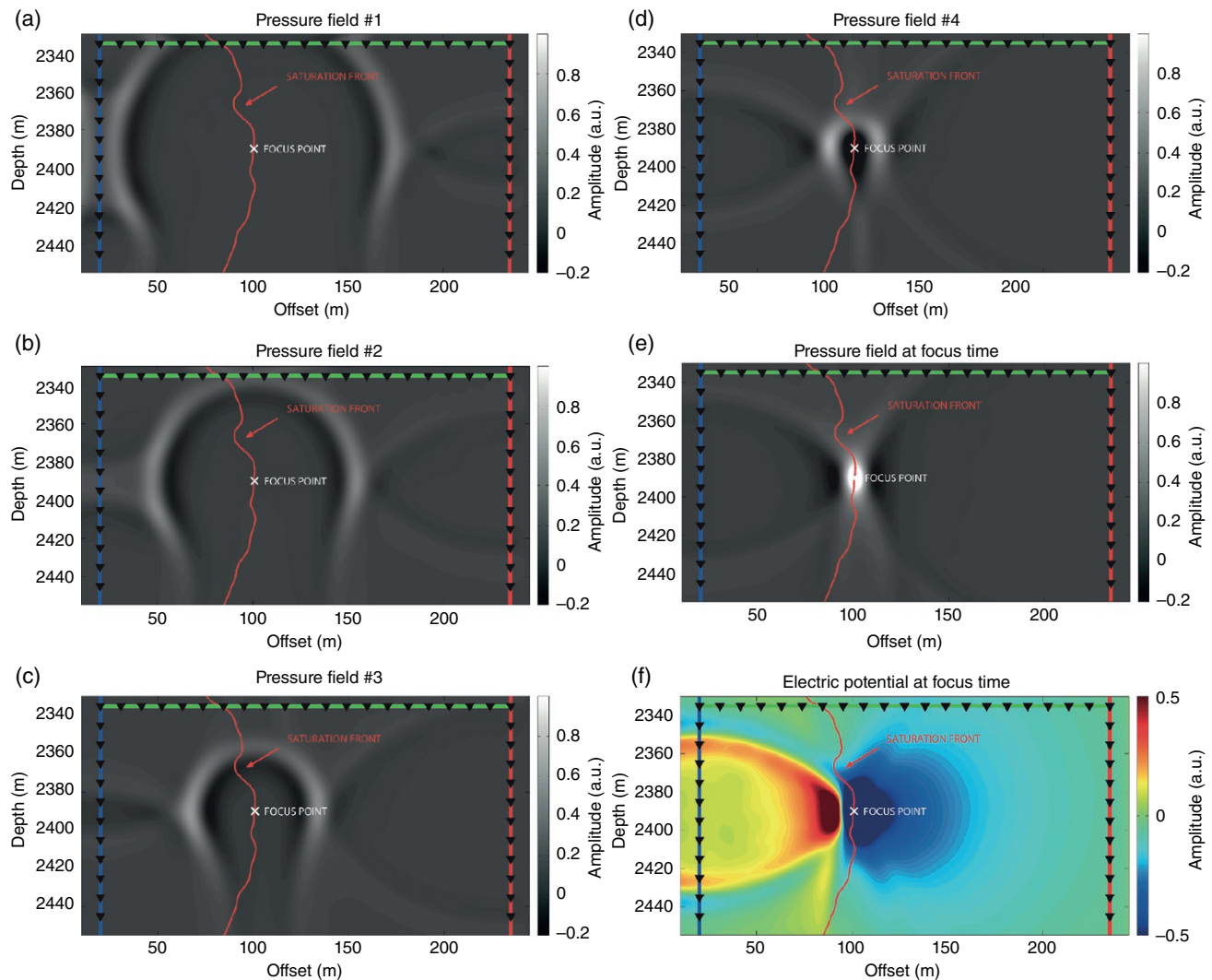


Figure 2.3 Seismic beamforming at the saturation front, and the resulting electrical potential distribution: (a–e) Evolution of the confining pressure $P(r, t)$ for five snapshots. The propagating wave fields interfere constructively at the focus point and focus time $t = 0.0$ s. (f) Electrical potential distribution corresponding to the seismic energy focused at the saturation front at the focus time (snapshot (e)).

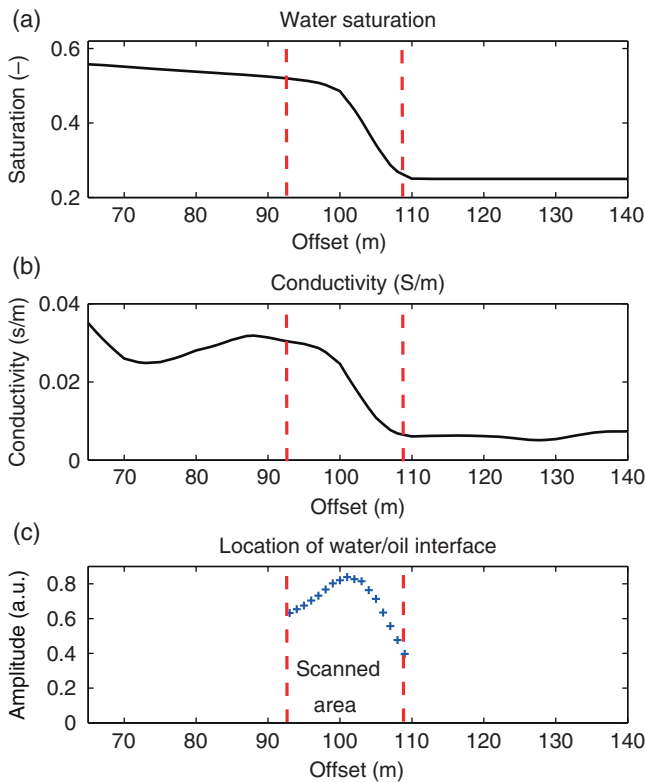


Figure 2.4 Determination of the position of the oil/water interface using a set of beamforming points at the same depth and crossing the position of the interface: (a) Spatial distribution of the water saturation showing the position of the saturation front, (b) electrical conductivity distribution, and (c) amplitude of the conversion as a function of offset. It can be observed that the strongest seismoelectric conversion occurs at the position of the saturation front, which is characterized by large electrical conductivity and saturation gradients.

introduced a range of theoretical formulations, from the fully coupled poroelastic-Maxwell equations, via a quasi-static approximation, where electromagnetic induction terms are neglected, to a poroacoustic formulation that describes the coupling of mechanical pressure waves to electric fields, using a minimum amount of porous medium parameters, which was subsequently adopted for seismoelectric beamforming, to scan media for contrasts of interest, and to maximize seismoelectric conversion.

We have presented a successful reciprocity test in a layered-Earth consisting of a free surface, a single layer, and a halfspace using the analytically based, numerical modeling code ESSEMOD. This example serves as a validation of the theoretical and numerical consistency of the theory presented in this chapter. We have illustrated the seismoelectric beamforming approach using a numerical example representing a water-flooding experiment containing varying degrees of saturation. This example shows

the success of this seismoelectric focusing approach in amplifying the seismoelectric conversion, as well as the sensitivity of the seismoelectric converted signals to wetting/nonwetting contrasts.

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